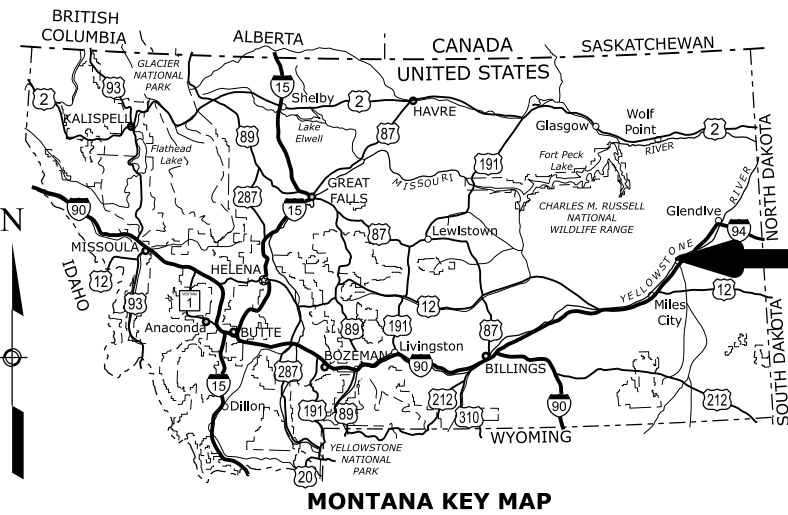


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U. S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION



PROJECT LOCATION

PLANS FOR PROPOSED PROJECT
MT PRAIRIE 2016(3)

MILWAUKEE BRIDGE RESURFACING
AND REHABILITATION

PRAIRIE COUNTY
MONTANA
LENGTH 0.348 MILES

TYPE OF CONSTRUCTION:

Bridge deck and bridge rail replacement,
pier repair, grading, and aggregate surfacing.

DESIGN DESIGNATION:

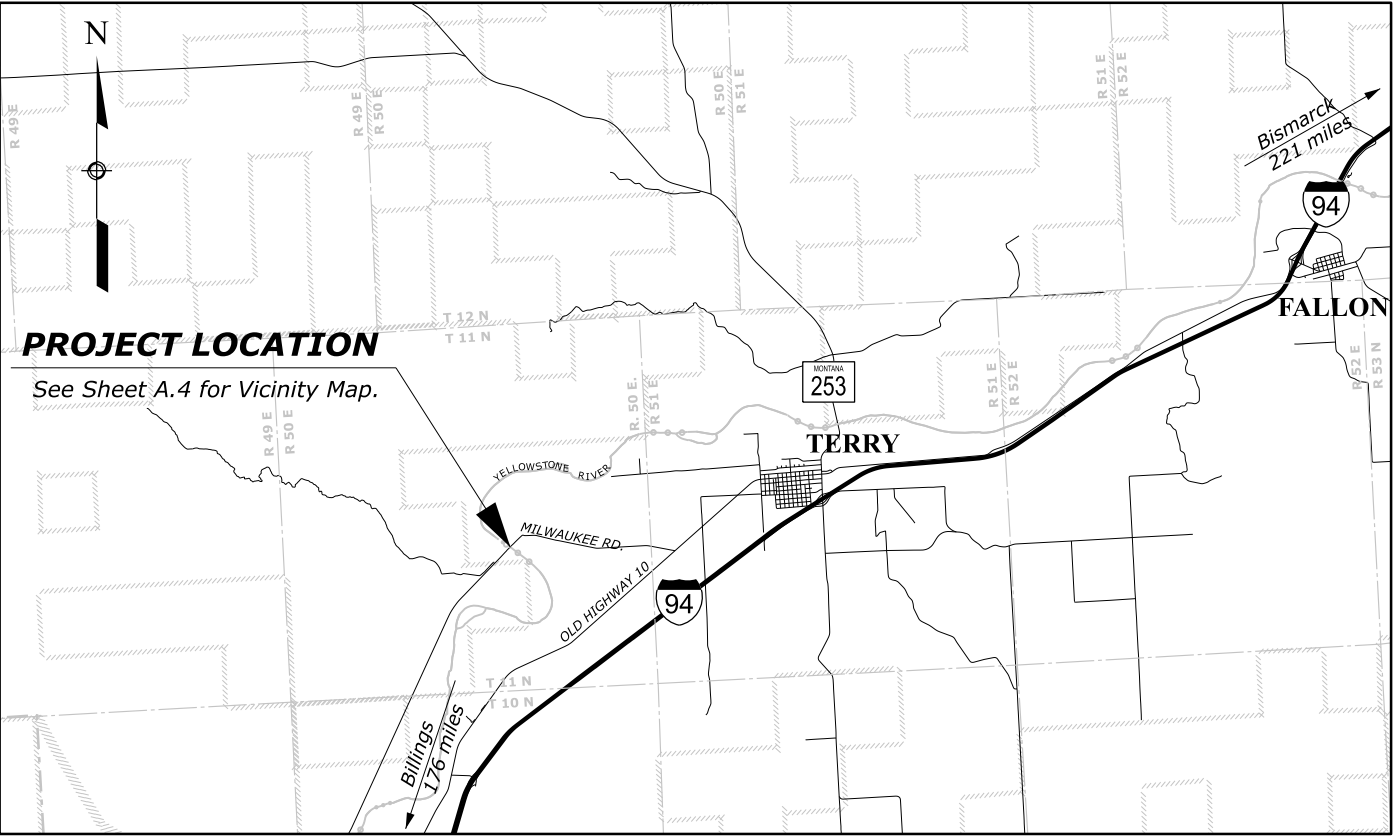
ADT (2025) <400
ADT (2045) <400
V 15 MPH
e (max) 0.060

SPECIFICATION:

Standard Specifications for
Construction of Roads and Bridges
on Federal Highway Projects, FP-14

PLANS PREPARED BY
U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
WESTERN FEDERAL LANDS HIGHWAY DIVISION
VANCOUVER, WASHINGTON

PROJECT MANAGER
J. KEY



PROJECT LOCATION
See Sheet A.4 for Vicinity Map.

SCALE OF MILES



PROJECT	SHEET NUMBER
MT PRAIRIE 2016(3)	A.1

SECTION INDEX

- A. GENERAL INFORMATION
- B. SUMMARY OF QUANTITIES
- C. TYPICAL SECTIONS
- D. PLAN-PROFILE
- E. APPROACH ROADS
- F. EROSION CONTROL
- G. HYDRAULICS
- H. (BLANK)
- I. (BLANK)
- J. TEMPORARY TRAFFIC CONTROL
- K. PERMANENT TRAFFIC CONTROL
- S. BRIDGE

See Sheet A.2 for complete Sheet Index.

PRELIMINARY
NOT FOR CONSTRUCTION

100%/FINAL
PLANS FOR QC

APPROVED:

Chief of Engineering,
Western Federal Lands Highway Division

DATE

c:\pw-work\0592470\mt-a-prairie\01603_aa.dgn [Sheet Index] 19 December 2025 2:37 PM

PROJECT	SHEET NUMBER
MT PRAIRIE 2016(3)	A.2

SHEET INDEX	
A. GENERAL INFORMATION	
A.1	TITLE SHEET
A.2	SHEET INDEX
A.3	PLAN SYMBOLS AND ABBREVIATIONS
A.4	VICINITY MAP
B. SUMMARY OF QUANTITIES	
B.1-2	SUMMARY OF QUANTITIES
C. TYPICAL SECTIONS	
C.1	TYPICAL SECTION
D. PLAN-PROFILE	
D.1	TABULATION OF PLAN QUANTITIES
D.2	EARTHWORK SUMMARY
D.3-6	PLAN-PROFILE
D.7	SURVEY CONTROL
E. APPROACH ROADS	
E.1	TABULATION OF APPROACH QUANTITIES
E.2	APPROACH ROAD PLAN-PROFILE
F. EROSION CONTROL	
F.1	TABULATION OF EROSION CONTROL QUANTITIES
F.2-3	EROSION CONTROL PLAN
F.4	STANDARD DETAIL W157-21
G. HYDRAULICS	
G.1	TABULATION OF HYDRAULICS QUANTITIES
G.2	PIER SCOUR COUNTERMEASURE TYPICAL SECTION
G.3	RIPRAP LAYOUT PIER #4 SCOUR PROTECTION
H. (BLANK)	

SHEET INDEX (continued)	
I. (BLANK)	
J. TEMPORARY TRAFFIC CONTROL	
J.1	TABULATION OF TEMPORARY TRAFFIC CONTROL QUANTITIES
J.2	STANDARD DETAIL W635-14
J.3	ROAD CLOSURE DETAIL
J.4	ADVANCED ROAD CLOSED SIGN DETAIL
K. PERMANENT TRAFFIC CONTROL	
K.1	TABULATION OF PERMANENT TRAFFIC CONTROL QUANTITIES
K.2-3	PERMANENT TRAFFIC CONTROL PLAN
K.4	STANDARD DETAIL W633-60

SHEET INDEX (continued)	
S. BRIDGE	
S.1	PLAN AND ELEVATION
S.2	GENERAL NOTES
S.3	SINGLE SPAN PLAN AND ELEVATION
S.4	TYPICAL SECTION
S.5	BENT CLEANUP PLAN
S.6	BENT IMPROVEMENTS
S.7	DECK FRAMING LAYOUT
S.8	TYPICAL SECTION DETAILS
S.9	DECK PANEL LAYOUT
S.10	DECK PANEL STUD SPACING
S.11	EXPANSION JOINT SPACING
S.12	RAIL POST DETAILS
S.13	RAIL POST CLEARANCE DETAILS

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Δ	total central angle
Δc	curve central angle
$\emptyset$	diameter
θs	spiral central angle
abut.	abutment
ADT	average daily traffic
AH	ahead
appr.	approach
BK	back
b.f.	back face
BM	bench mark
BP	balance point
br.	bridge
brg.	bearing
btwn	between
cc or c. to c.	center to center
cl.	centerline
clr.	clear
CMP	corrugated metal pipe
col.	column
conc.	concrete
conn.	connection
constr. jt.	construction joint
cont.	continuous
CS	point of curve to spiral
ctrs.	centers
CUFT	cubic foot (feet)
culv.	culvert
CUYD	cubic yard(s)
D	diameter
DHV	design hourly volume
dia.	diameter
diaph.	diaphragm
dist.	distance
drwg(s).	drawing(s)
E	east
e	superelevation rate
e.f.	each face
El. 94.16	elevation in feet
elev.	elevation
emb.	embankment
EOP	edge of pavement
EQ or eq.	equation
EW	edge of water
exc.	excavation
exp. jt.	expansion joint
f.f.	front face
fin.	finish
flg.	flange
ftg.	footing
ga.	gage (gauge)
galv.	galvanized
ID	inside diameter
IE	invert elevation
jt.	joint
KSI	thousand pounds per square inch
L	length of curve
lat.	latitude
LNFT	linear foot (feet)
long.	longitudinal
LPSM	lump sum
Ls	length of spiral
lt. or LT	left
LW	low water

M.L.	main line
M.P.	mile post
matl.	material
max.	maximum
MGAL	thousand gallon
min.	minimum
mon.	monument
N	north
NC	normal crown
o.c.	on center
o. to o.	out to out
OD	outside diameter
OG	original ground
OHWM	ordinary high water mark
PC	point of curve
PCC	point of compound curve
PCS	point of curve to spiral
PI	point of intersection
pl.	plate
POC	point on curve
POS	point on spiral
POT	point on tangent
PS	point of tangent to spiral
PSF	pounds per square foot
PSI	pounds per square inch
PSC	point of spiral to curve
PST	point of spiral to tangent
PT	point of tangent
pvmt.	pavement
R	radius
R.	range
R/W	right-of-way
rdwy.	roadway
reinf.	reinforcement
reqd.	required
rt. or RT	right
rte.	route
S	south
SADT	seasonal average daily traffic
SC	section
sect.	section
shldr.	shoulder
SLRY	slurry unit
spa.	spacing, spaces or spaced
SQFT	square foot
SQYD	square yard
SRS	point of spiral to reverse spiral
SS	point of spiral to spiral (no curve)
ST	point of spiral to tangent
STA, Sta.	station
std.	standard
stgr.	stringer
stiff.	stiffener
struc.	structural
STS	point of spiral to tangent spiral
T	tangent distance
T.	township
TBM	temporary bench mark
thd.	thread
TS	point of tangent to spiral
Ts	tangent distance (spiraled curve)
typ.	typical
V	design speed (velocity)
vph	vehicles per hour
VPI	vertical point of intersection
W	west

National Boundary
State Boundary
County Boundary
City Boundary
Township or Range Line

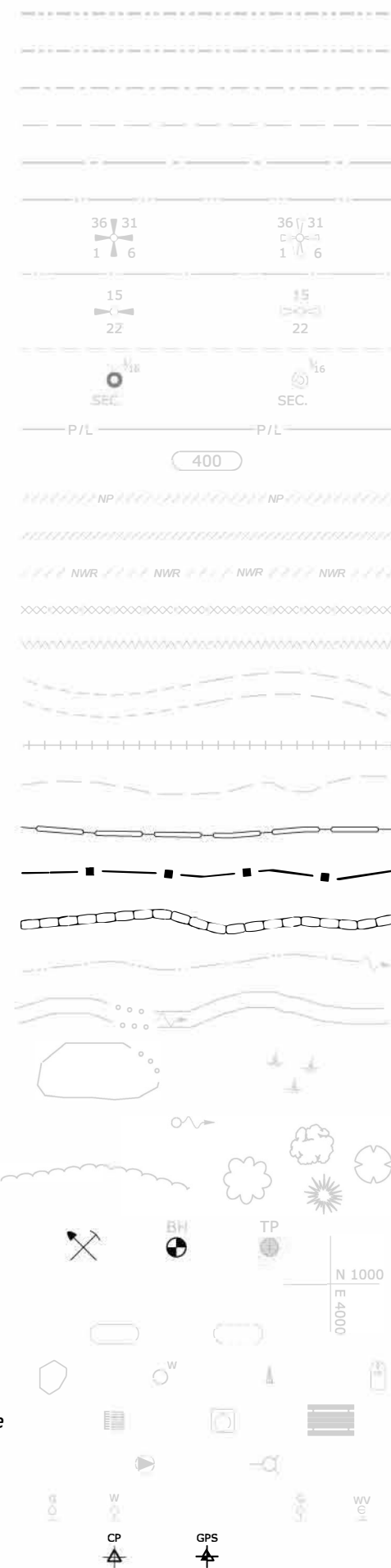
Section Line
Section Corner (Found, Projected)
 $\frac{1}{4}$ Section Line
 $\frac{1}{4}$ Section Corner (Found, Projected)
 $\frac{1}{16}$ Section Line
 $\frac{1}{16}$ Section Corner (Found, Projected)

Property Line
Parcel Number
National Park Boundary
National Forest Boundary
National Wildlife Refuge Boundary
BLM Lands Boundary
Indian Reservation Boundary
Existing Roadway (Paved, Gravel)

Railroad
Trail
Fiber Roll
Silt Fence
Sandbag
Intermittent Drainage or Small Creek
Large Creek or River

Lake, Pond or Reservoir; Marshland
Spring or Seep
Treeline; Individual Trees

Material Source; Bore Hole; Test Pit
Spot Elevation; Coordinate Grid Tick
Above Ground Tank; Underground Tank
Boulder; Well; Antenna; Grave
Cooking Grate; Garbage Can; Picnic Table
Flagpole; Fire Hydrant
Gas & Water Meter; Gas & Water Valve
Control Point (Terrestrial and GPS)



North Arrow



EXISTING

PROPOSED

Slope Stake Limits

Top of Cut
Toe of Fill

Construction Limits

- no symbol -

Bottom of Ditch

Fence

Gate with Fence

Cattleguard

Guardrail

Concrete Barrier and Guard Wall

Retaining Wall

wall face

Signs (single, double post; portable)

Delineators

Pipe Culvert (arrow shows flow)

Pipe Culvert with End Section

Pipe Culvert with Headwall

Pipe Culvert with Drop Inlet

Box Culvert

Underdrain

Overhead/Above Ground Utilities

Underground Utilities

FM = force main, FO = fiber optic, G = gas, IRR = irrigation, O = oil,
P = power, SA = sanitary sewer, SD = storm drain, SS = storm sewer,
STEAM = steam, T = telephone, TV = CATV, W = water

Poles (Power, Telephone, Joint Use,
Light, Support w/Anchor)

Miscellaneous Utility Features

EM = electric meter, TP = telephone pedestal, TV = CATV pedestal,
UP = transformer or junction box, WF = water fountain

Building

Right-of-Way Line

Permanent Easement

Construction Easement

- no symbol -

Riprap

Section A-A

NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION, FHWA
OFFICE OF FEDERAL LANDS HIGHWAY

**PLAN SYMBOLS AND
ABBREVIATIONS**

WFLHD DETAIL
W101-1

SPECIFICATION
FP-14

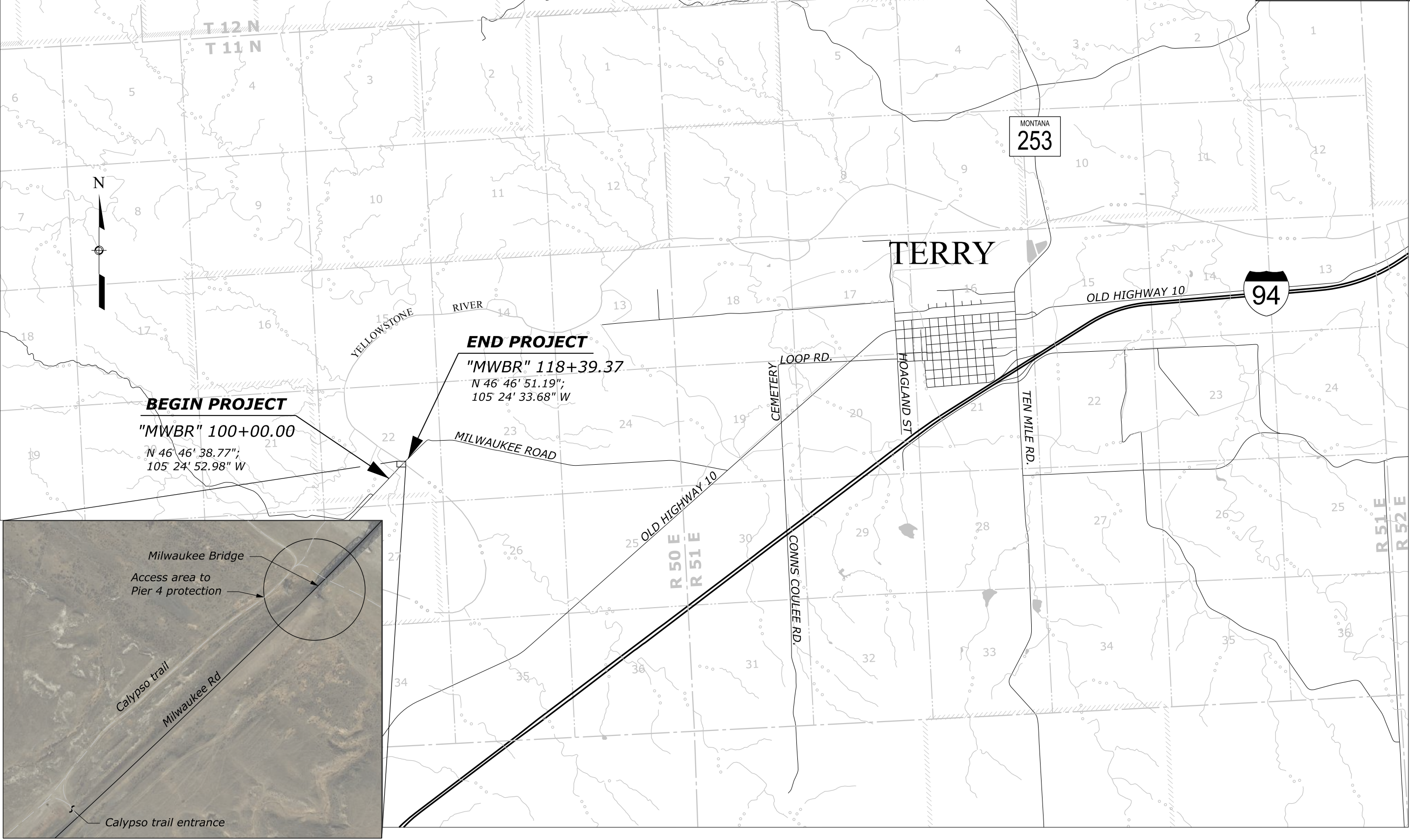
APPROVED FOR USE
6/2022

NOTE:

- Other symbols used in the plans will be shown in a legend on the appropriate plan sheet.

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PROJECT	SHEET NUMBER
MT PRAIRIE 2016(3)	A.4



VICINITY MAP

SUMMARY OF QUANTITIES

PROJECT

SHEET NUMBER

MT PRAIRIE 2016(3)

B.1

A M E N D	LINE ITEM NO.	PAY ITEM NUMBER	PAY ITEM DESCRIPTION	UNIT	Sheet and Description								ALLOW- ANCE	Estimated Quantities	Remarks and/or Determination of Estimated Quantity
					SECTION D	SECTION E	SECTION F	SECTION G	SECTION I	SECTION J	SECTION K	SECTION S		BID SCHEDULE	
					PLAN & PROFILE	APPROACH ROADS	EROSION CONTROL	HYDRAULI CS	MISCELLA NEOUS	TEMPORAR Y TRAFFIC CONTROL	PERMANEN T TRAFFIC CONTROL	STRUCTUR ES			
	A0020	15101-0000	MOBILIZATION	LPSM										ALL	
	A0040	15201-0000	CONSTRUCTION SURVEY AND STAKING	LPSM										ALL	
	A0060	15301-0000	CONTRACTOR QUALITY CONTROL	LPSM										ALL	
	A0080	15401-0000	CONTRACTOR TESTING	LPSM										ALL	
	A0100	15501-0000	CONSTRUCTION SCHEDULE	LPSM										ALL	
	A0120	15703-1500	SOIL EROSION CONTROL, TEMPORARY SOIL TACKIFIER	ACRE			0.8						0.1	0.9	
	A0140	15705-1400	SOIL EROSION CONTROL, FIBER ROLL	LNFT			1498						102	1600	
	A0160	15802-0000	WATERING FOR DUST CONTROL	LPSM										ALL	
	A0180	20101-0000	CLEARING AND GRUBBING	ACRE	0.2	0.1							0.1	0.4	
	A0200	20301-2300	REMOVAL OF SIGN/MARKER	EACH							4			4	
	A0220	20301-2400	REMOVAL OF SIGN	EACH							2			2	
	A0240	20302-0150	REMOVAL OF BRIDGE RAILING	LNFT								2160		2160	
	A0260	20303-0200	REMOVAL OF BRIDGE DECK	SQYD								1520		1520	
	A0280	20401-0000	ROADWAY EXCAVATION	CUYD	93	4							3	100	
	A0300	20402-0000	SUBEXCAVATION	CUYD				1849					91	1940	
	A0320	20403-0000	UNCLASSIFIED BORROW	CUYD	413								37	450	
	A0340	20815-0000	COFFERDAMS	LPSM										ALL	
	A0360	25101-0500	PLACED RIPRAP, METHOD A, CLASS 5	CUYD				1900					100	2000	
	A0380	30201-2000	ROADWAY AGGREGATE, METHOD 2	CUYD	589	18							33	640	
	A0400	55222-0000	REPAIR CONCRETE,(BENT IMPROVEMENT)	LPSM										ALL	
	A0420	55502-0000	STRUCTURAL STEEL, FURNISHED, FABRICATED, AND ERECTED	LB								382000		382000	
	A0440	55601-0900	BRIDGE RAILING, STEEL	LNFT								2170		2170	
	A0460	56202-0000	TEMPORARY SUPPORT STRUCTURE,(WORK PLATFORM)	LPSM										ALL	
	A0480	56320-0000	CONTAINMENT SYSTEM AND WORKER PROTECTION PLAN	LPSM										ALL	
	A0500	61921-1000	REMOVE AND RESET FENCE	LNFT	53								7	60	
	A0520	62201-0250	DUMP TRUCK, 10 CUBIC YARD MINIMUM CAPACITY	HOURL										40	
	A0540	62201-0600	BACKHOE LOADER, 8 CUBIC FOOT MINIMUM RATED CAPACITY BUCKET, 30-INCH WIDTH	HOURL										20	
	A0560	62201-0850	WHEEL LOADER, 1 CUBIC YARD MINIMUM RATED CAPACITY	HOURL										20	
	A0580	62201-2800	MOTOR GRADER, 8 FOOT MINIMUM BLADE	HOURL										20	
	A0600	62201-3400	HYDRAULIC EXCAVATOR, 1 CUBIC YARD MINIMUM CAPACITY WITH THUMB ATTACHMENT	HOURL										40	
	A0620	62201-3750	CHAIN SAW	HOURL										10	
	A0640	62301-0000	GENERAL LABOR	HOURL										40	
	A0660	62402-0300	FURNISHING AND PLACING TOPSOIL, 4-INCH DEPTH	ACRE	0.3	0.1								0.4	
	A0680	62501-0000	TURF ESTABLISHMENT	ACRE	0.3	0.1							0.1	0.5	
	A0700	63302-0000	SIGN SYSTEM	SQFT							31.5		2.5	34	

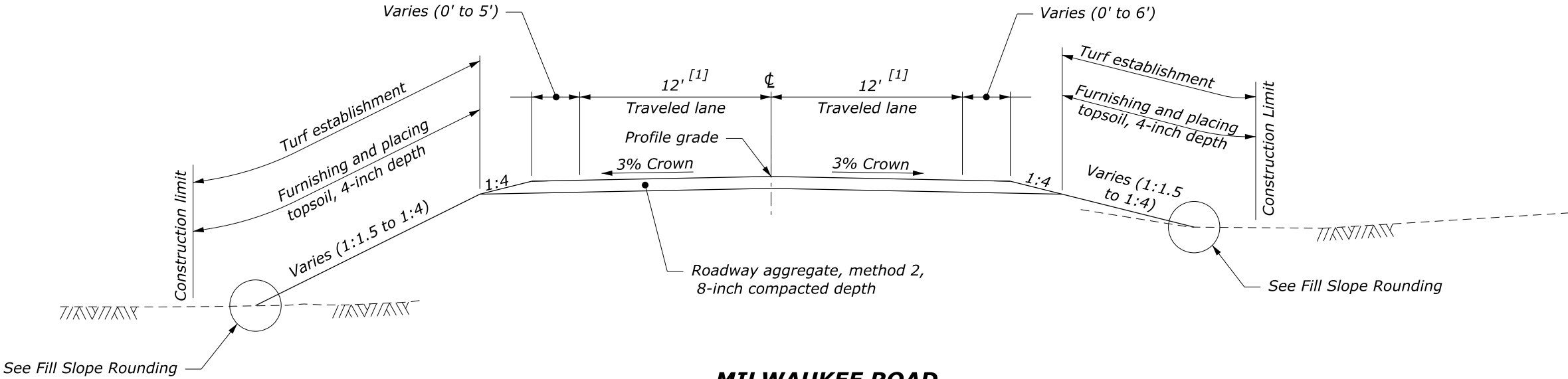
Milestone: 100% Estimate; Date Completed:

SUMMARY OF QUANTITIES

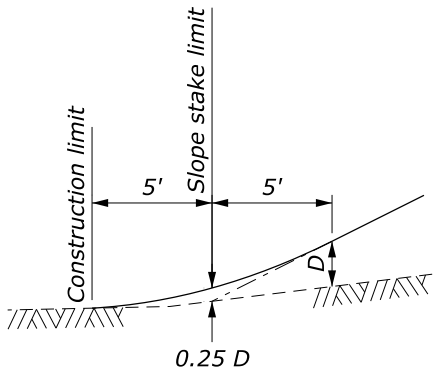
PROJECT	SHEET NUMBER
MT PRAIRIE 2016(3)	B.2

A M E N D	LINE ITEM NO.	PAY ITEM NUMBER	PAY ITEM DESCRIPTION	UNIT	Sheet and Description								ALLOW- ANCE	Estimated Quantities	Remarks and/or Determination of Estimated Quantity
					SECTION D	SECTION E	SECTION F	SECTION G	SECTION I	SECTION J	SECTION K	SECTION S		BID SCHEDULE	
					PLAN & PROFILE	APPROACH ROADS	EROSION CONTROL	HYDRAULI CS	MISCELLA NEOUS	TEMPORAR Y TRAFFIC CONTROL	PERMANEN T TRAFFIC CONTROL	STRUCTUR ES			
	A0720	63308-0000	OBJECT MARKER	EACH							4			4	
	A0740	63309-0000	DELINEATOR	EACH							2		2	4	
	A0760	63502-0600	TEMPORARY TRAFFIC CONTROL, BARRICADE TYPE 3	EACH						6			1	7	
	A0780	63502-1300	TEMPORARY TRAFFIC CONTROL, DRUM	EACH						20			2	22	
	A0800	63504-1000	TEMPORARY TRAFFIC CONTROL, CONSTRUCTION SIGN	SQFT						189			11	200	
	A0820	63506-0500	TEMPORARY TRAFFIC CONTROL, FLAGGER	HOURL						200				200	

Milestone: 100% Estimate; Date Completed



MILWAUKEE ROAD
100+00.00 to 103+93.31
114+75.12 to 118+39.24



FILL SLOPE ROUNDING

FOOTNOTE:
[1] Travel lane width varies from 7' to 12' at bridge approach transitions.

NO SCALE

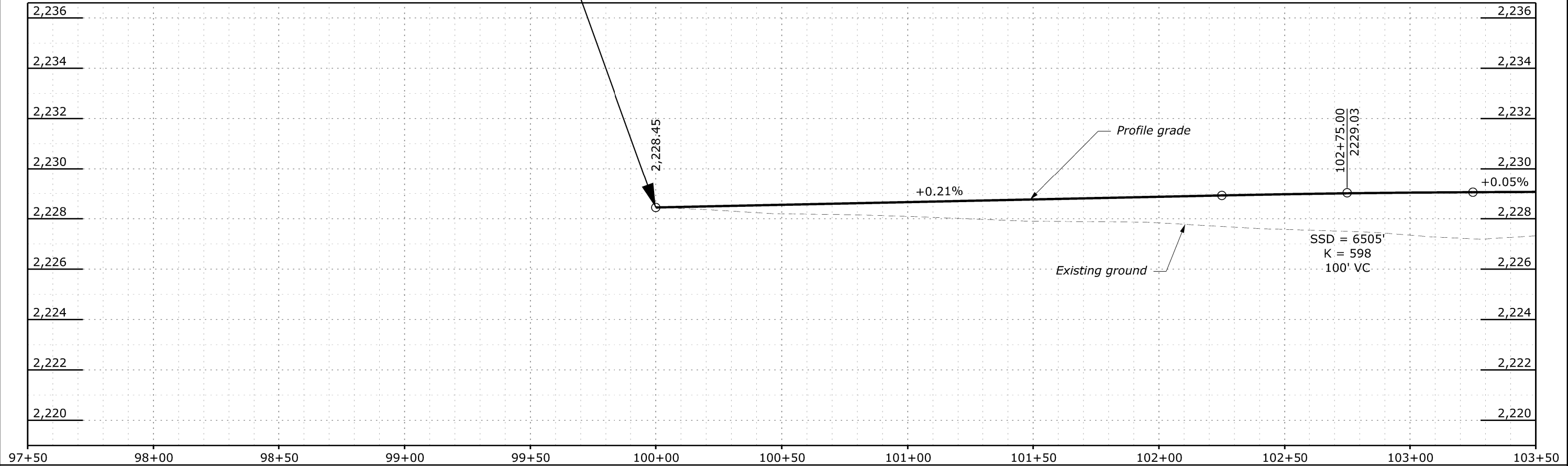
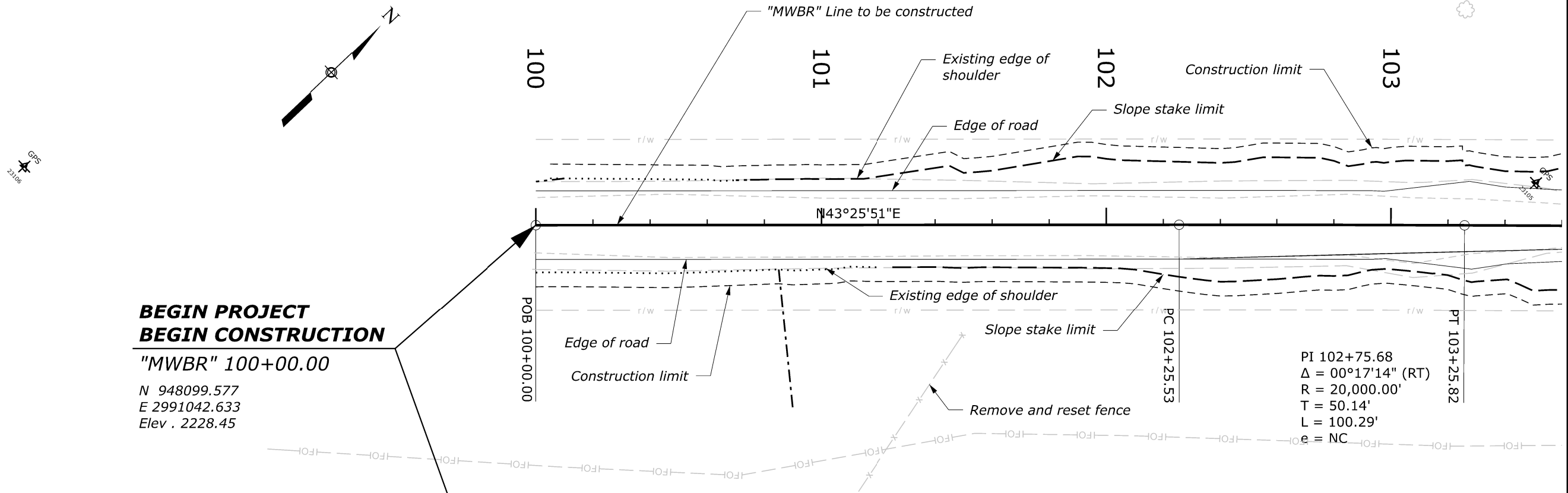
TYPICAL SECTION

PLAN QUANTITIES				
ITEM	DESCRIPTION	UNIT	QUANTITIES	NOTES
20101-0000	CLEARING AND GRUBBING	ACRE	0.2	
30201-2000	ROADWAY AGGREGATE, METHOD 2	CUYD	589	
61901-1000	REMOVE AND RESET FENCE	LNFT	53	Remove and reset portions of fence within construction limit.
62402-0300	FURNISHING AND PLACING TOPSOIL, 4-INCH DEPTH	ACRE	0.3	
62515-1000	TURF ESTABLISHMENT	ACRE	0.3	

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EARTHWORK SUMMARY TABLE													
LOCATION	CUT						FILL				TOTALS		
	Item 20401-0000 ROADWAY EXCAVATION	(+) Additional Excavation (See Note 2) for info only	(-) Topsoil Stripped from Cuts for info only	(-) Estimate of Unsuitable Material for info only	Total Unadjusted Excavation Available for Fill for info only	Total Adjusted Available for Fill (adjusted based on shrink/swell factor) for info only	Embankment for info only	(+) Additional Backfill Needed (See Note 3) for info only	(+) Topsoil stripped from embankment foundation areas for info only	Total Fill Needed for info only	Total Adjusted Available For Fill less Total Fill Needed for info only	Waste for info only	Item 20403-0000 UNCLASSIFIED BORROW
UNITS	CUYD	CUYD	CUYD	CUYD	CUYD	CUYD	CUYD	CUYD	CUYD	CUYD	CUYD	CUYD	CUYD
Mainline (MWBR)	93		23	-	70	60	321		143	465	-405		
Approach (BLAP)	4		-	-	4	3	12		-	12	-8		
TOTALS	97	0	23	0	74	63	333	0	143	476	-413	0	413

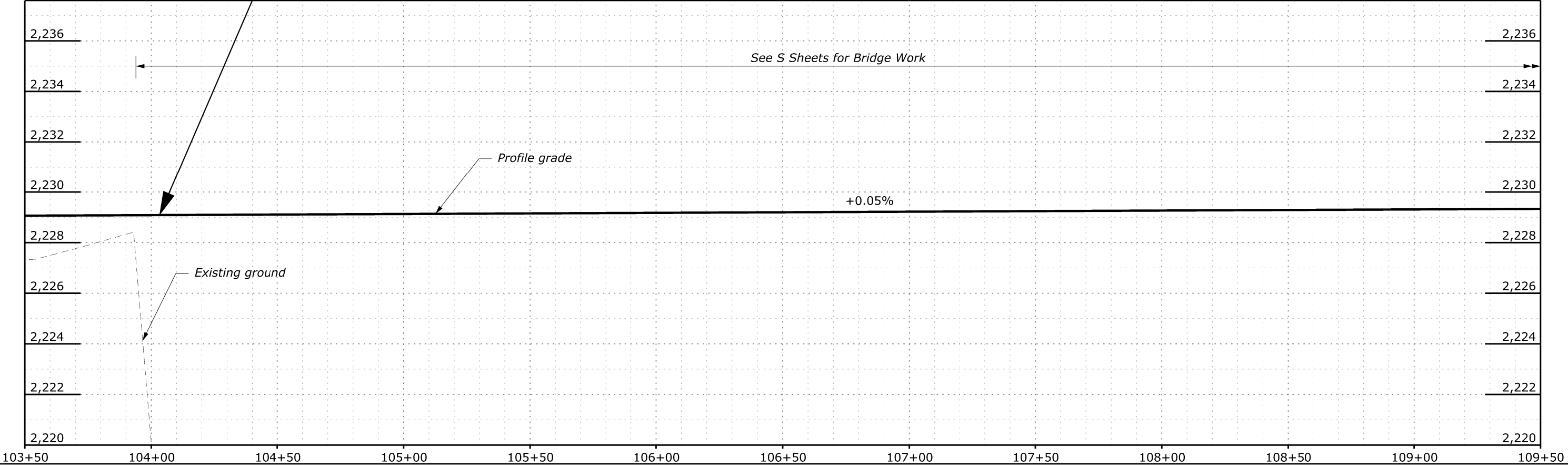
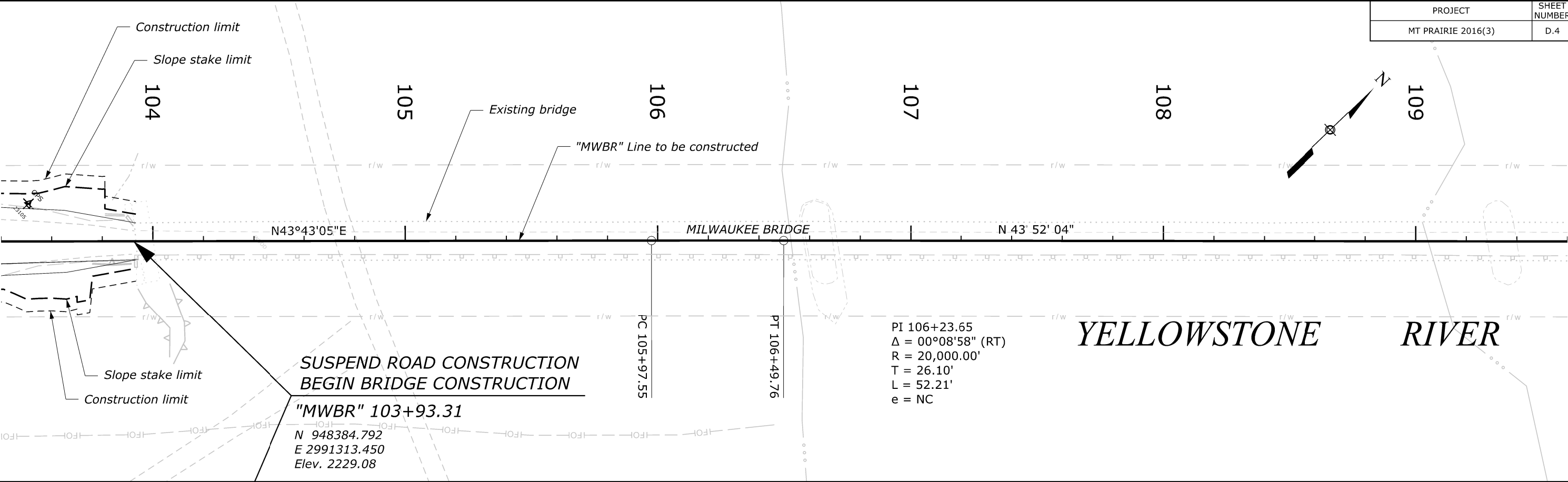
- NOTE:
1. All volumes are in-place cubic yards (i.e. in situ or compacted in place).
 2. Additional excavation that is suitable for use in embankment construction (Structural Excavation).
 3. Additional material needed for embankment construction (Structural backfill).



**PLAN-PROFILE
100+00 TO 103+50**

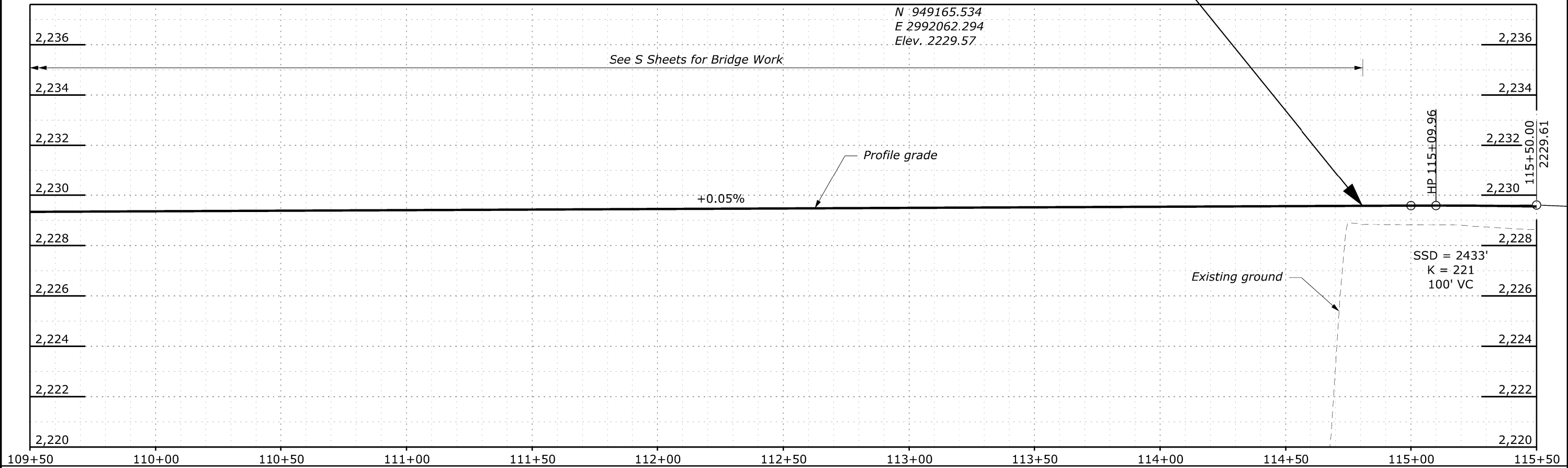
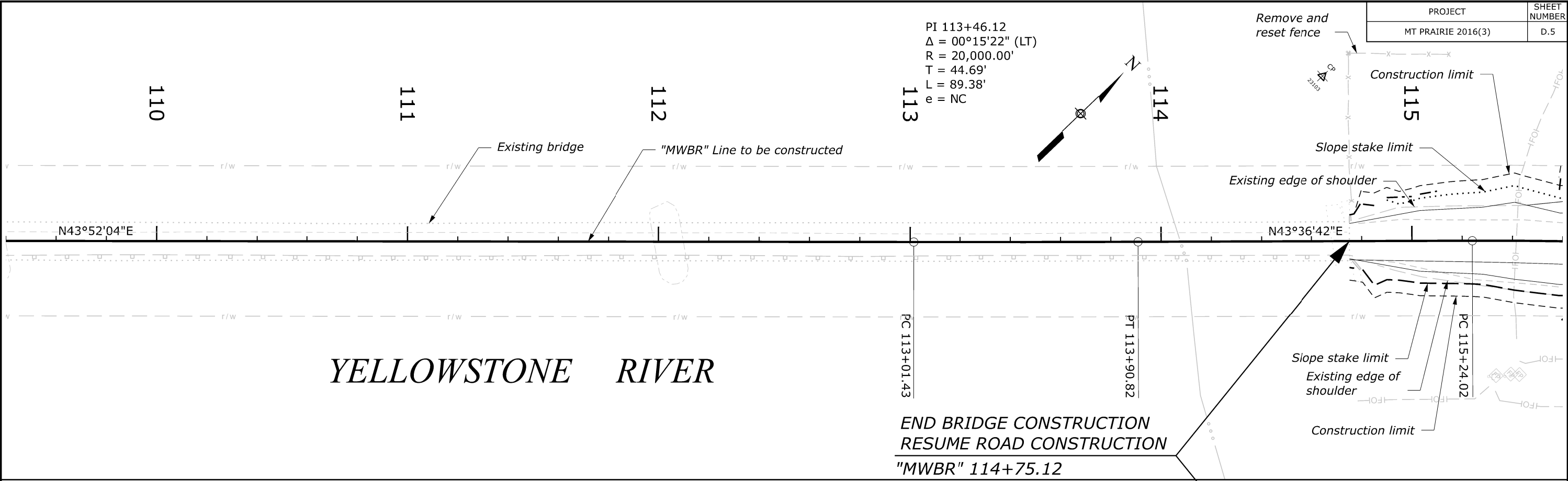
c:\pw-wor\0592473\mt-a-pra\201603_fb.dgn [MWBR - Plan 2 (Sheet)] 19 December 2025 9:58 AM

PROJECT	SHEET NUMBER
MT PRAIRIE 2016(3)	D.4

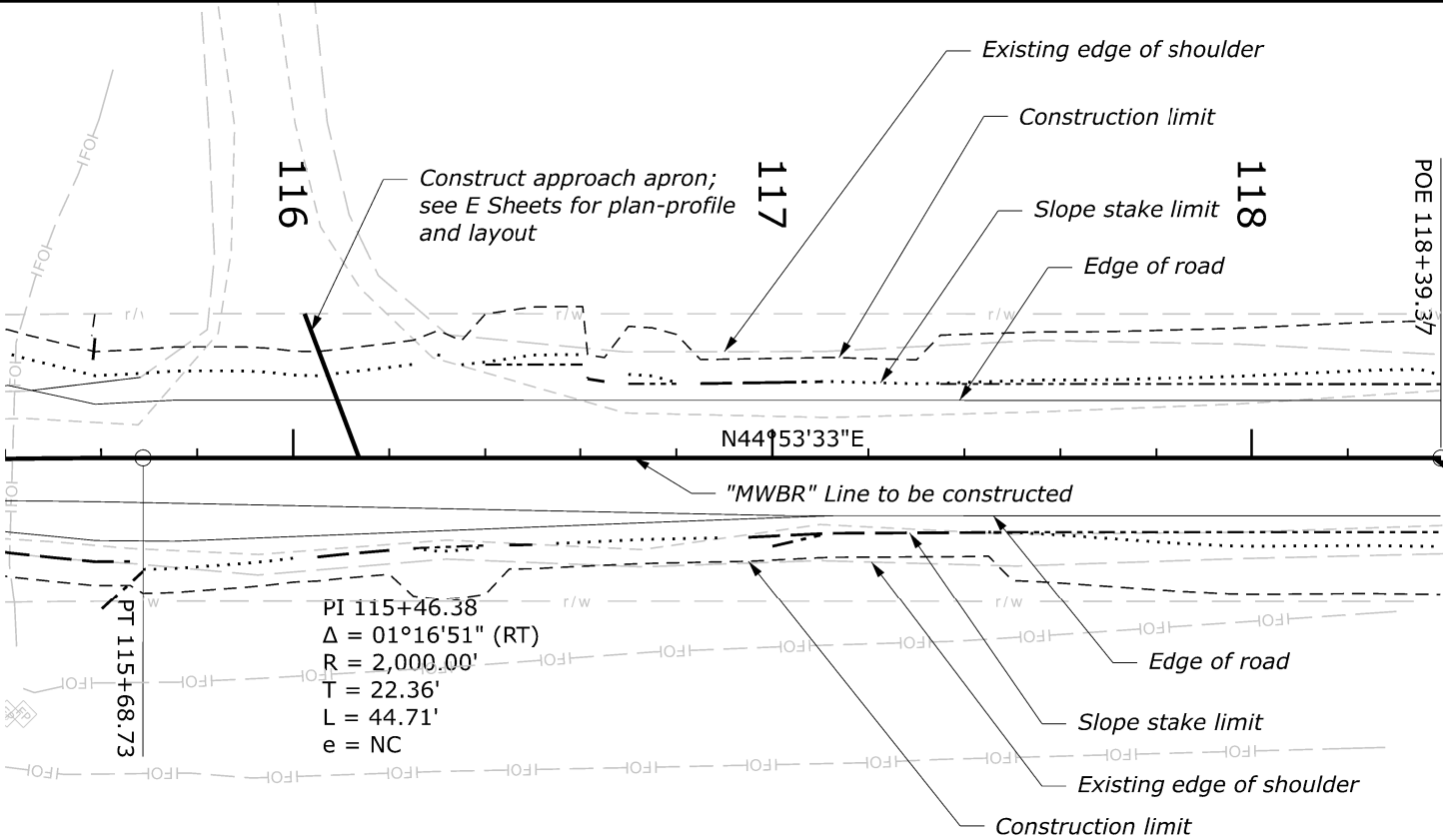


**PLAN-PROFILE
103+50 TO 109+50**

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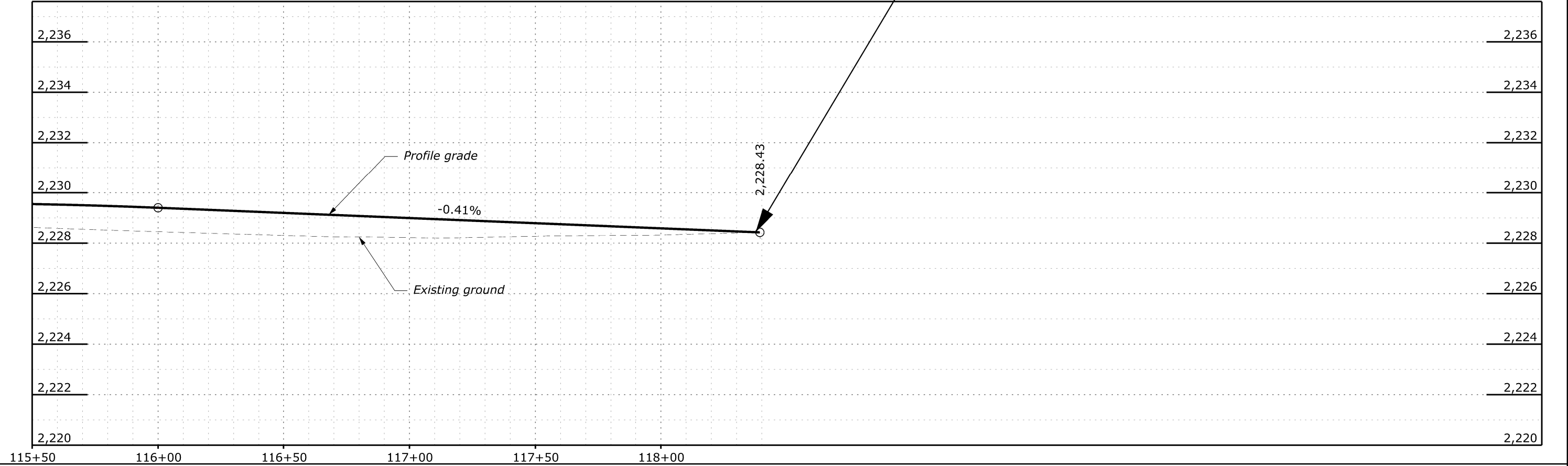


**PLAN-PROFILE
109+50 TO 115+50**



**END PROJECT
END CONSTRUCTION**

"MWBR" 118+39.24
N 949424.690
E 2992318.232
Elev. 2228.43



**PLAN-PROFILE
115+50 TO 118+39.24**

19-Dec-2025 9:49

c:\pw-work\0592473\Milwaukee_Bridge_controlsheets.xlsx]Sheet

Designed by:

WFLHD_Survey

--/----

Checked by:

WFLHD_Survey

--/----

PROJECT :Milwaukee Bridge Resurfacing and Rehabilitation

DATE OF FIELD WORK :June 2023

DATE OF FINAL ADJUSTMENT : 6/14/2023

PROJECT UNITS : International Foot

COORDINATE SYSTEM :Montana SPCS NAD83 2011

VERTICAL DATUM : Orthometric elevations based on the NAVD88 GEOID18

EPOCH DATE : 2010.0000

CADD COORDINATE SYSTEM : MT83/2011IF NSRS11(NAD83/2011) Montana, Intl Foot

POINT NUMBER	STATE PLANE COORDINATES			GEO COORDINATES				DESCRIPTION
	NORTH	EAST	ELEVATION	LATITUDE	LONGITUDE	ELLIPSOID HEIGHT	COMBINED FACTOR	
23101	950002.376	2992866.273	2228.449	46°46'56.621070"N	105°24'25.375112"W	2175.1325	0.999296818	5/8" ir w/fhwa alum cap
23102	949613.926	2992471.541	2227.801	46°46'52.993366"N	105°24'31.334288"W	2174.4886	0.999296919	5/8" ir w/fhwa alum cap
23103	949203.205	2992007.539	2227.297	46°46'49.181615"N	105°24'38.304495"W	2173.9896	0.999297017	5/8" ir w/fhwa alum cap
23104	948640.979	2991309.905	2191.675	46°46'43.995779"N	105°24'48.742322"W	2138.3743	0.999298819	5/8" ir w/fhwa alum cap
23105	948364.341	2991273.326	2226.697	46°46'41.286119"N	105°24'49.474879"W	2173.3981	0.9992972	5/8" ir w/fhwa alum cap
23106	947983.270	2990904.130	2228.385	46°46'37.717699"N	105°24'55.060985"W	2175.0897	0.99929719	5/8" ir w/fhwa alum cap

NOTE:

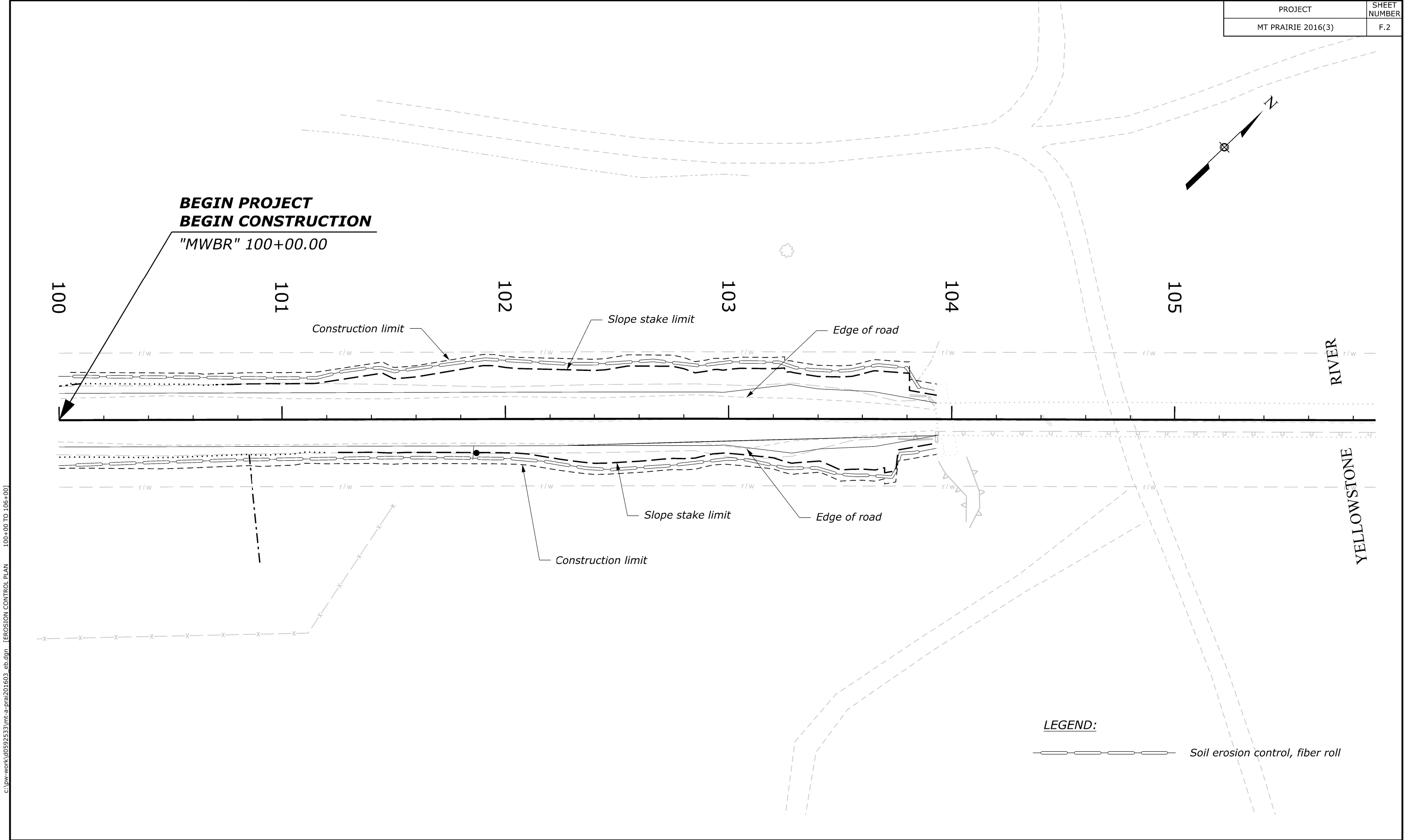
1. To precisely check distances between points as measured on the ground, inverse the state plane coordinates and divide the computed distance by a mean combined factor of the two points.

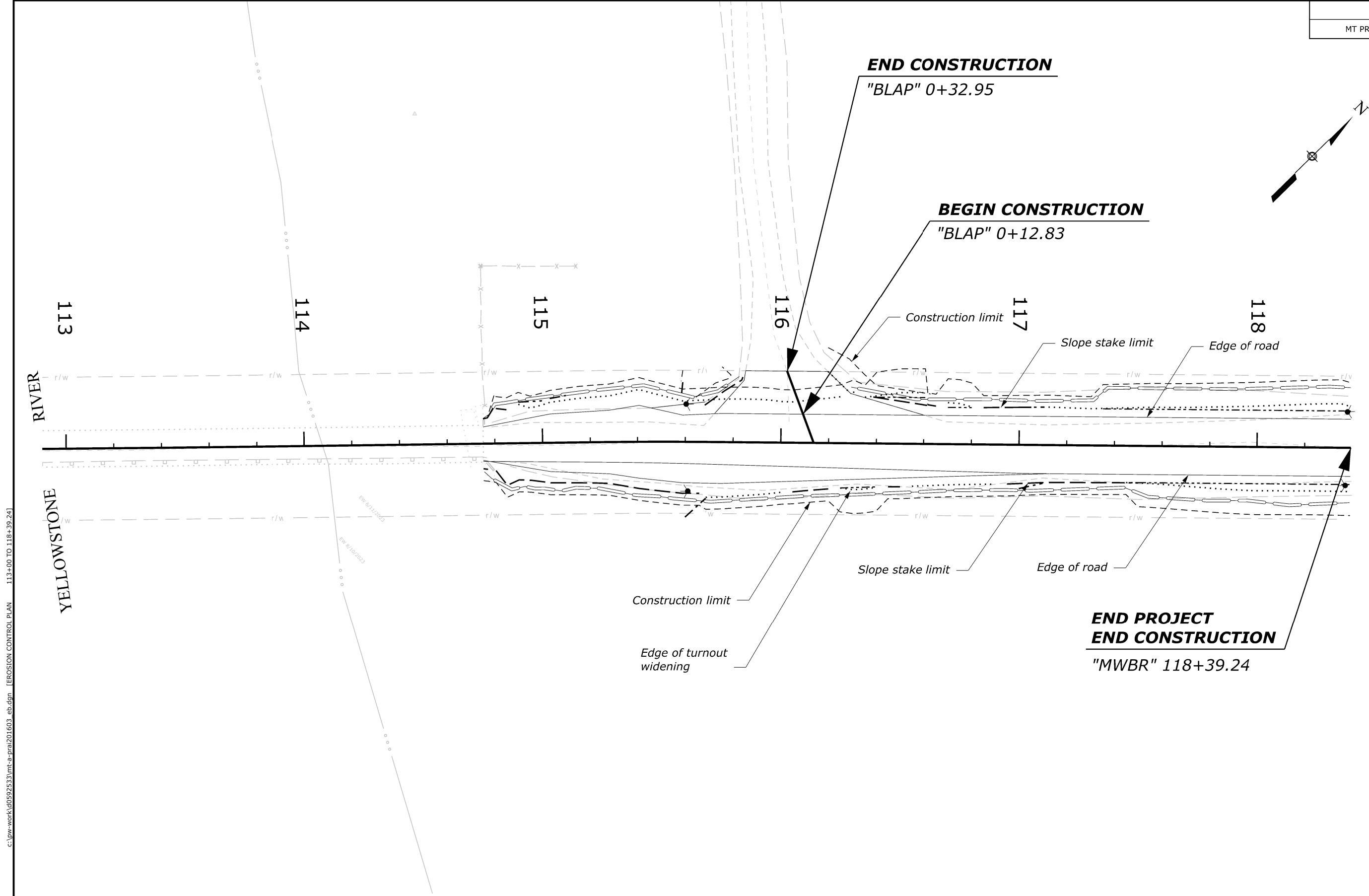
SURVEY CONTROL

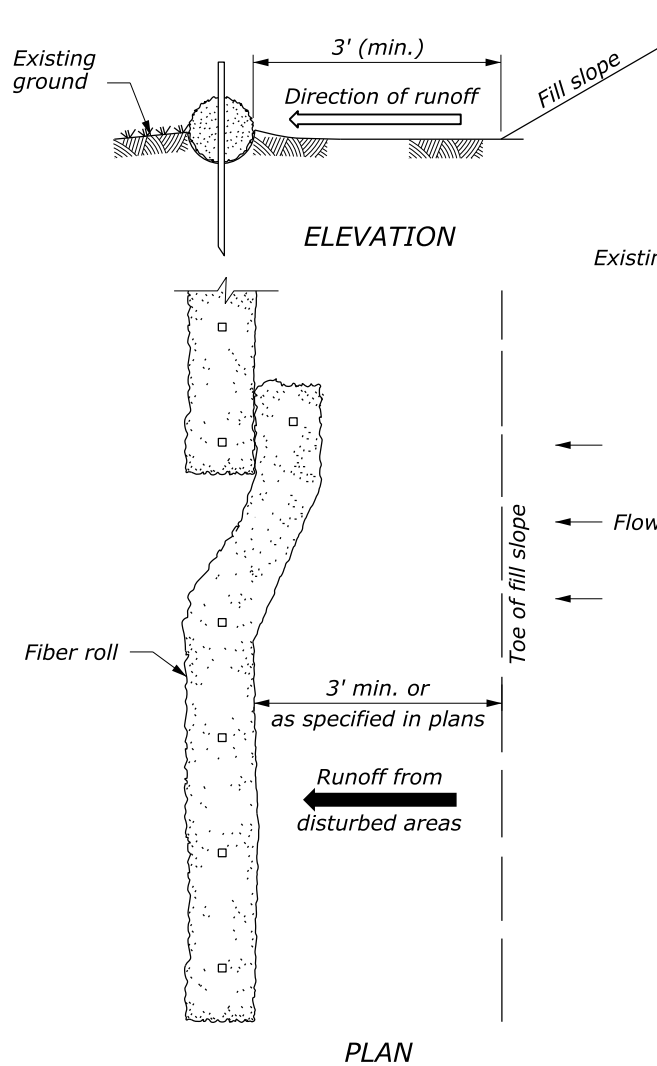
APPROACH QUANTITIES			
ITEM	DESCRIPTION	UNIT	QUANTITIES
20101-0000	CLEARING AND GRUBBING	ACRE	0.1
30201-2000	ROADWAY AGGREGATE, METHOD 2	CUYD	18
62406-0300	PLACING CONSERVED TOPSOIL, 4-INCH DEPTH	ACRE	0.1
62515-1000	TURF ESTABLISHMENT	ACRE	0.1

- NOTE:**
- See Sheet D.2, Earthwork Summary, for Roadway Excavation and Unclassified Borrow Quantities.

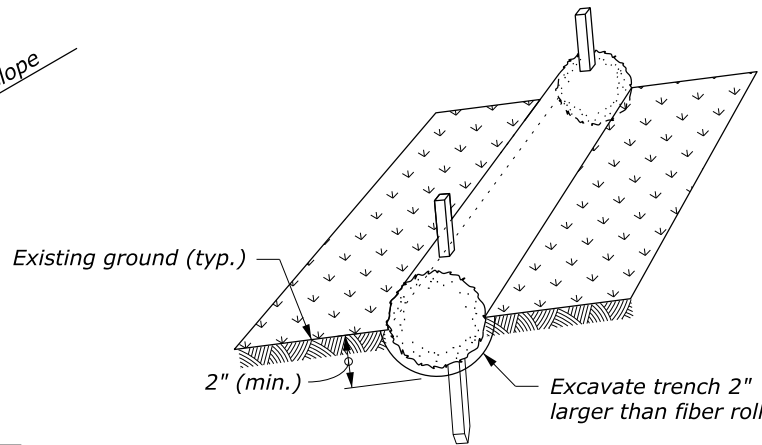
EROSION CONTROL QUANTITIES					
ITEM	DESCRIPTION	UNIT	LOCATIONS		TOTAL
			MAINLINE	APPROACH	
15703-1500	SOIL EROSION CONTROL, TEMPORARY SOIL TACKIFIER	ACRE	0.7	0.1	0.8
15705-1400	SOIL EROSION CONTROL, FIBER ROLL	LNFT	1,498	0	1,498



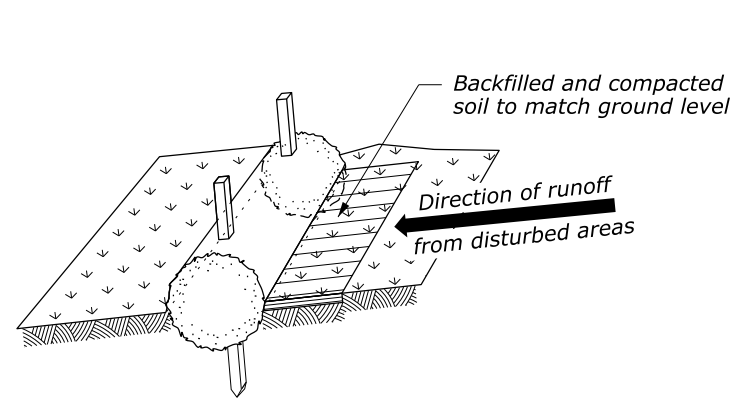




INSTALLATION BEYOND TOE OF SLOPE



Step 1: Excavate trench and install fiber rolls

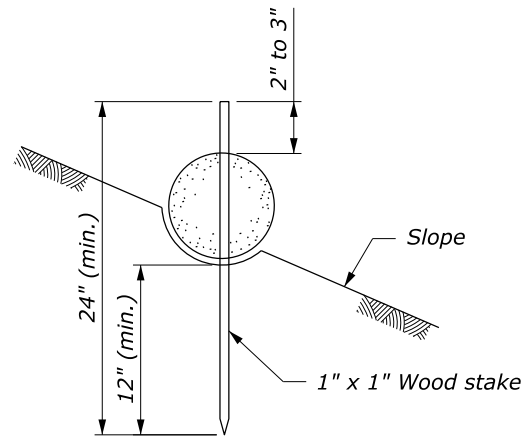


Step 2: Backfill soil against fiber rolls

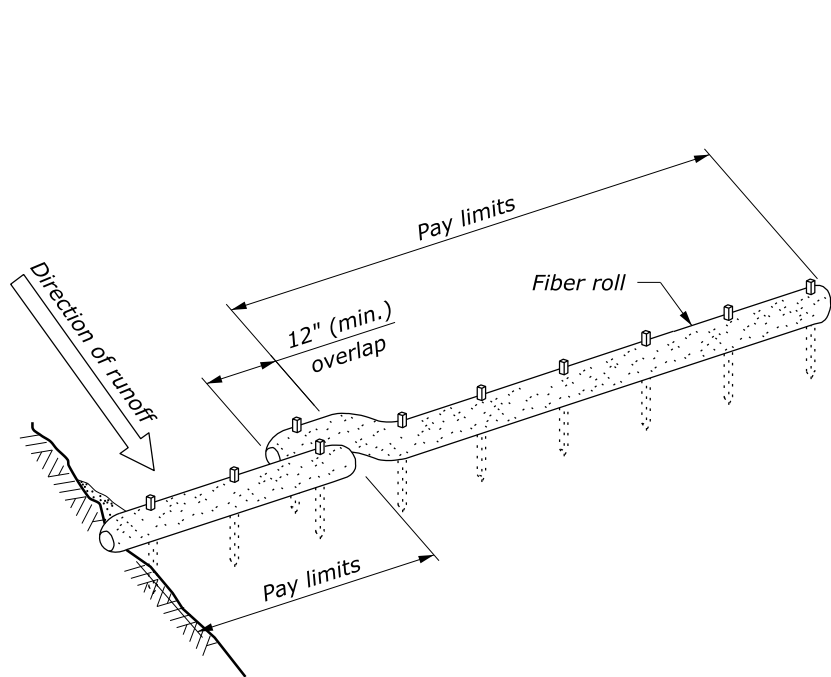
PROPERLY STAKED AND ENTRENCHED FIBER ROLL

FIBER ROLL SPACING	
Slope	Spacing (FT)
1:4 or flatter	40
1:3	30
1:2	20
1:1	10

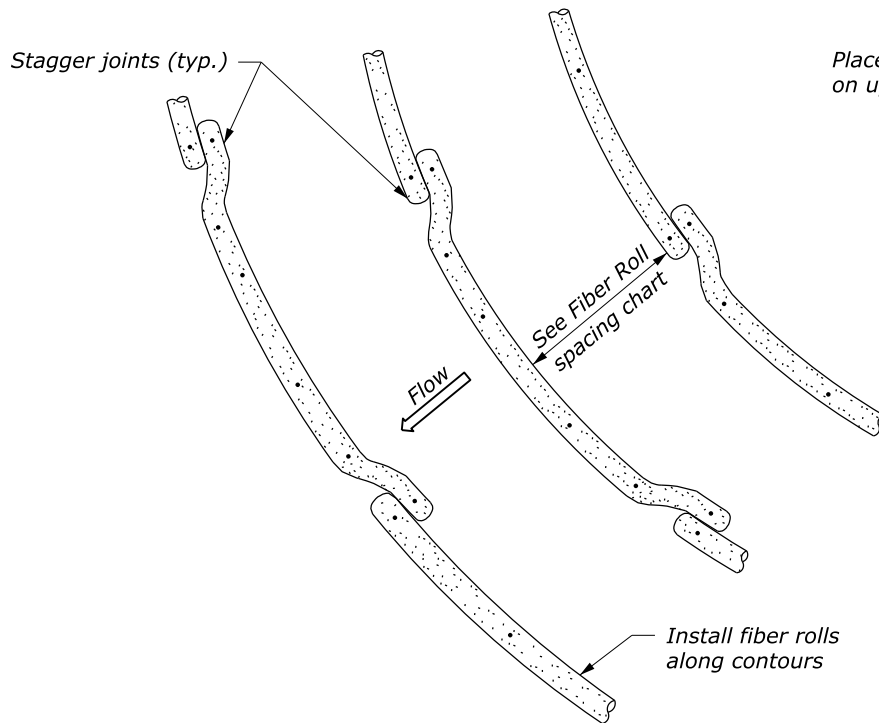
STAKES REQUIRED	
Fiber roll length (FT)	Stakes required for each roll
25	8
20	6
12	4



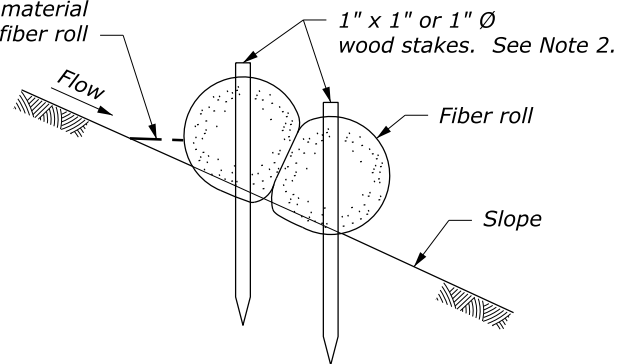
FIBER ROLL STAKING DETAIL



**ALTERNATE FIBER ROLL JOINT DETAIL
SLOPE PROTECTION INSTALLATION**



INSTALLATION ALONG SLOPES



FIBER ROLL LAPPING DETAIL

NO SCALE

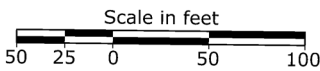
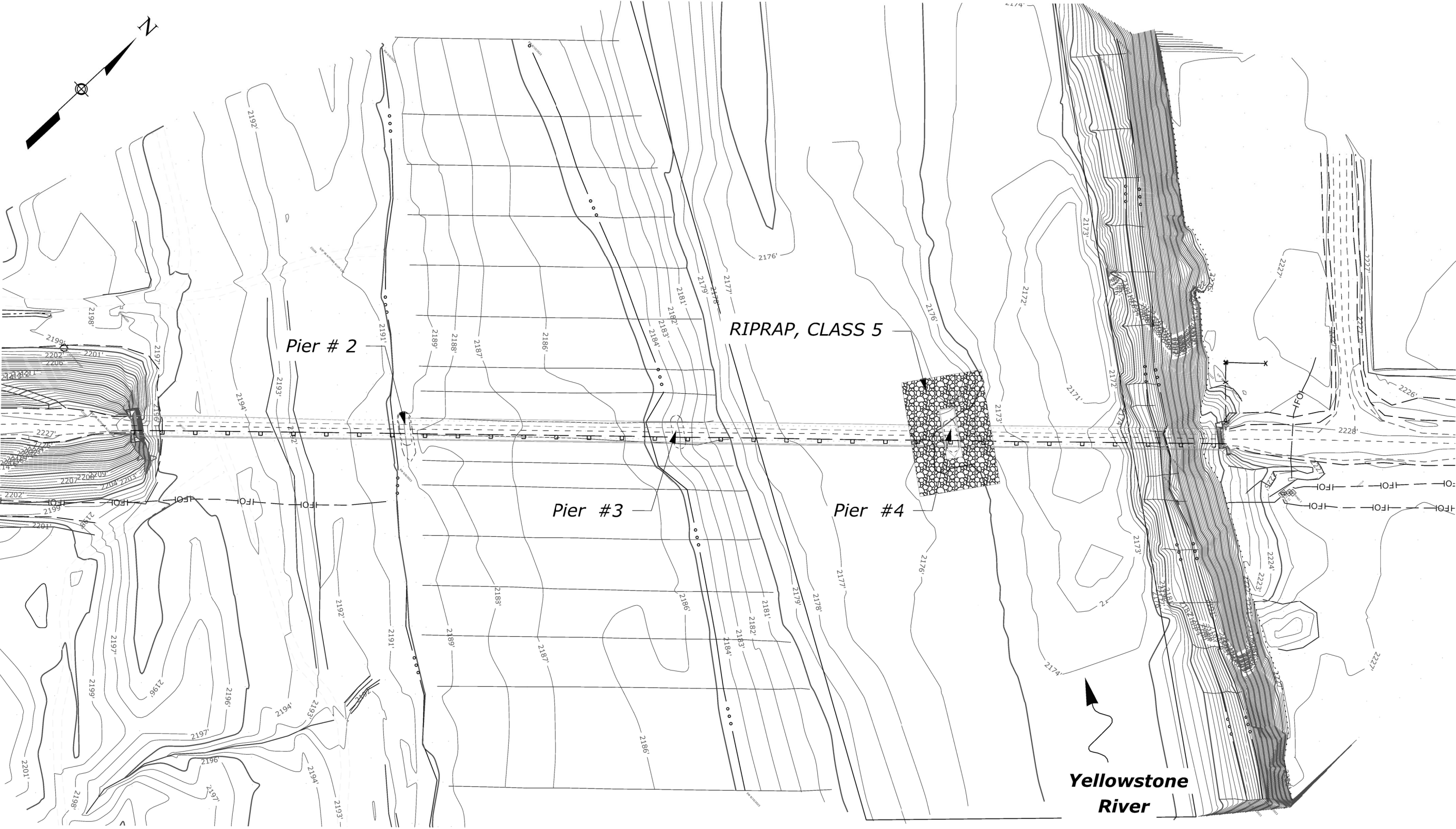
c:\pw-work\0592534\mt-a-pra\201603_ha.dgn [TABULATION OF HYDRAULICS QUANTITIES] 3 September 2025 8:52 AM

PROJECT	SHEET NUMBER
MT PRAIRIE 2016(3)	G.1

HYDRAULIC QUANTITIES			
ITEM NUMBER	DESCRIPTION	UNIT	QUANTITY
20815-0000	COFFERDAMS	LPSM	ALL
25101-0500	PLACED RIPRAP, CLASS 5	CUYD	1900
20402-0000	SUBEXCAVATION	CUYD	1940

INFORMATIONAL QUANTITIES ONLY		
DESCRIPTION	UNIT	QUANTITY
GEOTEXTILE FILTER, CLASS 1, NON-WOVEN	SQYD	575

TABULATION OF
HYDRAULICS QUANTITIES



**RIPRAP LAYOUT PIER #4
SCOUR PROTECTION**

c:\pw-work\0592537\mt-a-prairie\01603.ta.dgn [TABULATION OF TEMPORARY TRAFFIC CONTROL QUANTITIES] 14 April 2025 2:54 PM

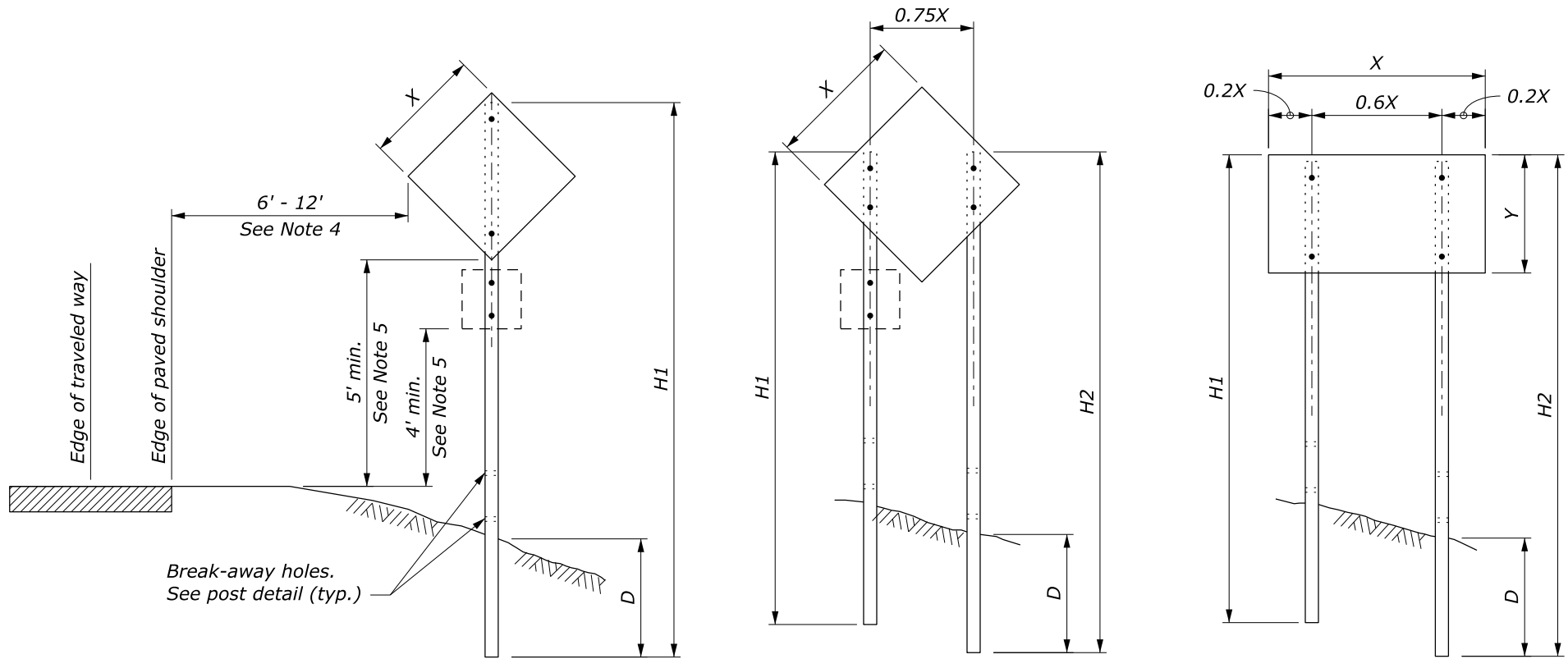
PROJECT	SHEET NUMBER
MT PRAIRIE 2016(3)	J.1

TEMPORARY TRAFFIC CONTROL DEVICE QUANTITIES			
ITEM NO.	DESCRIPTION	UNIT	QUANTITY
63502-0600	TEMPORARY TRAFFIC CONTROL, BARRICADE TYPE 3	EACH	6
63502-1300	TEMPORARY TRAFFIC CONTROL, DRUM	EACH	20
63506-0500	TEMPORARY TRAFFIC CONTROL, FLAGGER	HOURL	16

ITEM 63504-1000 TEMPORARY TRAFFIC CONTROL, CONSTRUCTION SIGN								
MUTCD NO.	SIGN MESSAGE	SIZE W X H (IN X IN)			AREA (SQFT)	NO. OF SIGNS	QUANTITY (SQFT)	Remarks
-----	SWPPP SIGN	36	x	42	10.50	2	21.00	Install at each end of the project corridor, facing oncoming traffic, as directed by the CO.
-----	REF-1	66	x	30	13.75	2	27.50	
-----	REF-2	66	x	30	13.75	2	27.50	
G20-2	"END ROAD WORK"	48	x	18	6.00	2	12.00	
R11-2	"ROAD CLOSED"	48	x	30	10.00	2	20.00	
W20-3	"ROAD CLOSED AHEAD"	36	x	36	9.00	2	18.00	
W20-4	"ONE LANE ROAD AHEAD"	36	x	36	9.00	2	18.00	
W13-1P	"15 MPH"	12	x	18	1.50	2	3.00	
W16-2P	"100 FEET"	18	x	12	1.50	2	3.00	
W16-P	"500 FEET"	18	x	12	1.50	2	3.00	
W20-1	"ROAD WORK AHEAD"	36	x	36	9.00	2	18.00	
W20-7	FLAGGER SYMBOL	36	x	36	9.00	2	18.00	
TOTAL							189.00	

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22 June 2024 11:17 AM

PROJECT	SHEET NUMBER
MT PRAIRIE 2016(3)	J.2



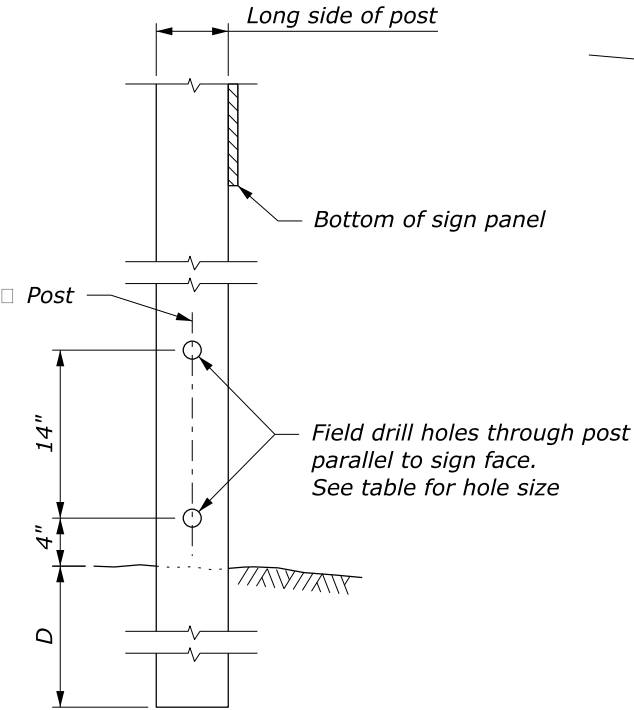
SINGLE POST SIGN

TWO POST SIGN

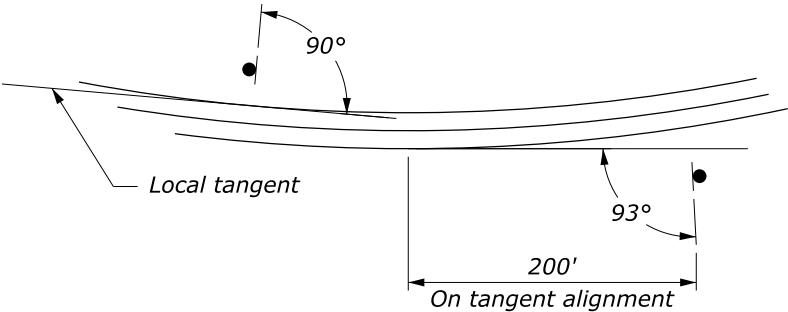
NOTE:

1. Attach sign panels with a minimum of 2 - 1/4" dia. bolts per post.
2. H1 and H2 = Overall post length.
Select post lengths to fit field conditions.
3. D = Post embedment depth for average soil conditions.
4. In areas where lateral distance is limited, a minimum lateral offset of 2' may be used. In areas with curbs, a minimum lateral distance of 1' behind the face of the curb may be used.
5. In pedestrian locations, or in areas with obstructed views, use 7' minimum mounting height for main sign and 6' minimum mounting height for secondary sign.
6. Use 7' minimum spacing between posts for sign posts 6" x 6" or larger.
7. State standards may be used as an alternative if approved by the CO.

WOOD POST SELECTION TABLE					
WIDTH "X"	AREA (SQFT)	NUMBER OF POSTS	POST SIZE (INCH)	D (INCH)	HOLE SIZE (INCH)
Diamond ≤ 36" Other Shapes ≤ 48"	< 10	1	4 x 4	36	0
		1	4 x 6	48	1.5
Diamond ≤ 48"	10 - 20	1	6 x 6	48	2
Diamond ≤ 48" Other Shapes ≤ 12'	10 - 20	2	4 x 4	36	0
	20 - 50	2	4 x 6	48	1.5
> 13'	50 - 65	2	6 x 6	48	2
12' - 16'	50 - 65	3	4 x 6	48	1.5
> 17'	65 - 95	4	4 x 6	48	1.5
> 30'	65 - 95	3	6 x 6	48	2



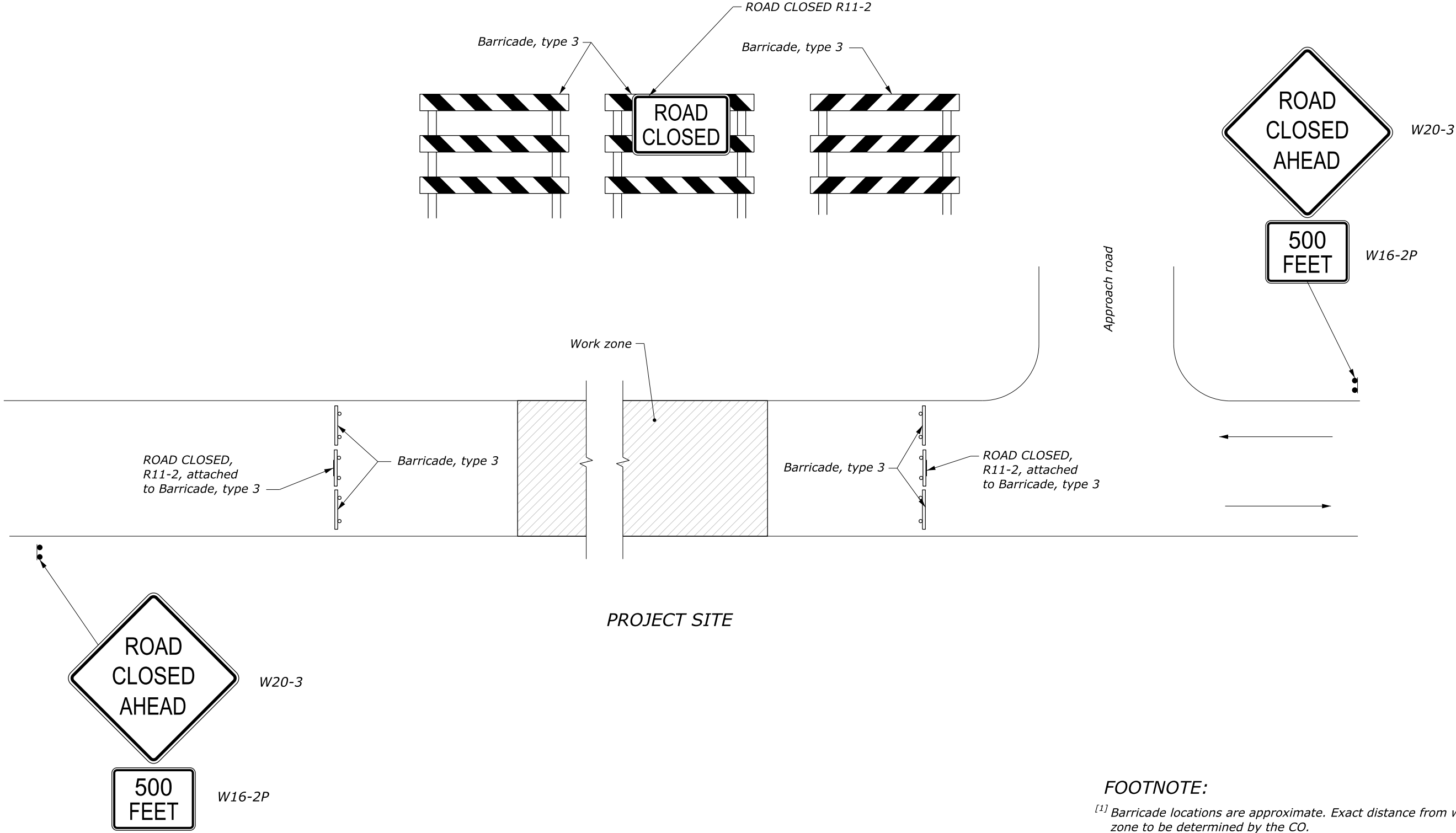
POST DETAIL



SIGN INSTALLATION ANGLE

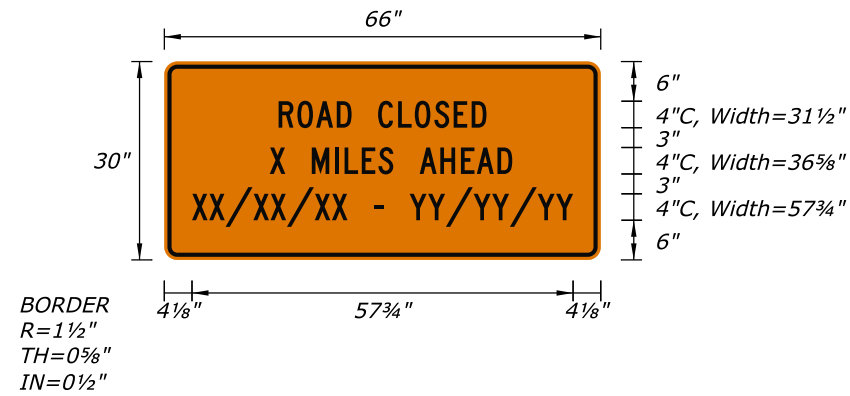
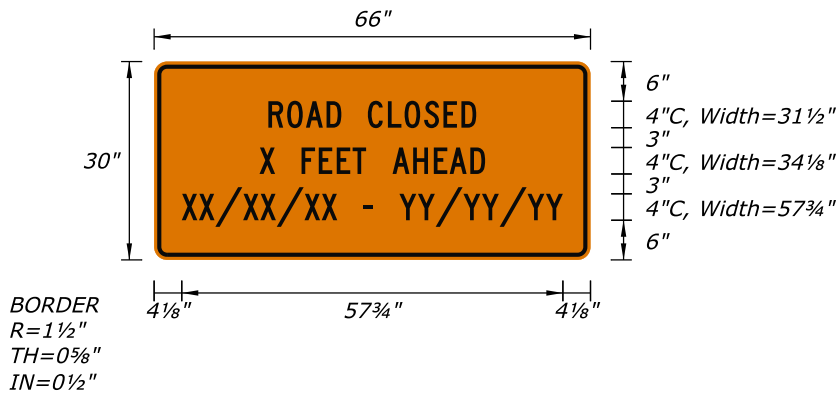
NO SCALE

U.S. DEPARTMENT OF TRANSPORTATION FEDERAL HIGHWAY ADMINISTRATION OFFICE OF FEDERAL LANDS HIGHWAY	
FLH STANDARD TEMPORARY TRAFFIC CONTROL SIGN INSTALLATION WOOD POSTS	
STANDARD APPROVED FOR USE 6/2005 REVISED: 7/2022	STANDARD 635-14



FOOTNOTE:
[1] Barricade locations are approximate. Exact distance from work zone to be determined by the CO.

ROAD CLOSURE DETAIL



NOTES:

1. Place according to the CO.

NO SCALE

**ADVANCED ROAD CLOSED
SIGN DETAIL**

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PROJECT	SHEET NUMBER
MT PRAIRIE 2016(3)	K.1

PERMANENT SIGN QUANTITIES										
ALIGNMENT	STATION	SIDE	MUTCD NO.	MESSAGE	NO. OF POSTS	PANEL DIMENSIONS W X H (IN. x IN.)			ITEM 20301-2400 REMOVAL OF SIGN (EACH)	ITEM 63302-0000 SIGN SYSTEM (SQFT)
"APPR"	10+92	LT	R1-1	"STOP"	1	36	x	36	---	9.00
"MWBR"	101+87	RT	W5-3	"ONE LANE BRIDGE"	1	36	x	36	1	9.00
			W13-1P	"15 MPH"		18	x	18		2.25
	102+87	RT	R1-2	"YIELD"	1	36	x	36	---	4.50
			R1-2aP	"TO ONCOMING TRAFFIC"		24	x	18		3.00
"MWBR"	117+45	LT	W5-3	"ONE LANE BRIDGE"	1	36	x	36	1	9.00
			W13-1P	"15 MPH"		18	x	18		2.25
	115+35	LT	R1-2	"YIELD"	1	36	x	36	---	4.50
			R1-2aP	"TO ONCOMING TRAFFIC"		24	x	18		3.00
TOTAL									2	39.00

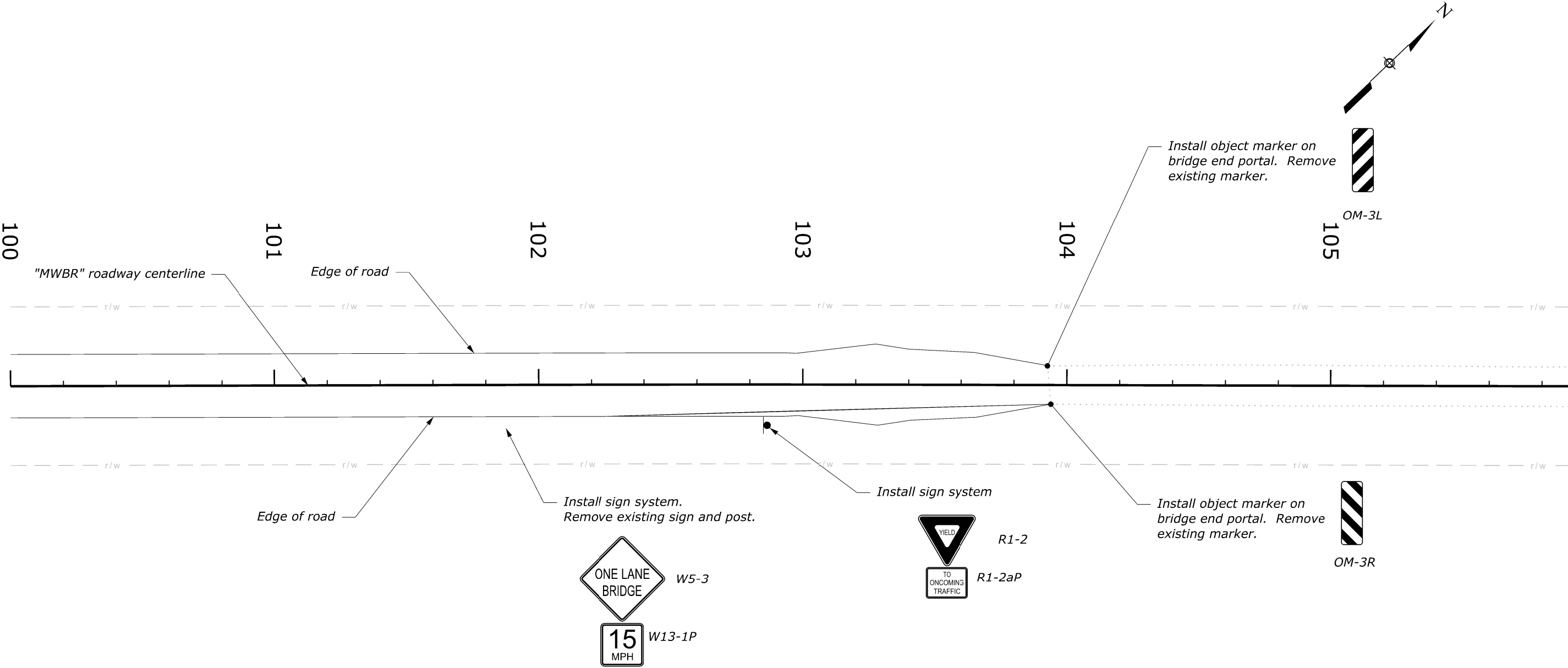
ITEM 63309-0000 DELINEATOR			
ALIGNMENT	STATION	SIDE	QUANTITY (EACH)
"MWBR"	115+60	LT	1
"MWBR"	115+60	RT	1
TOTAL			2

OBJECT MARKER QUANTITIES					
ALIGNMENT	STATION	SIDE	MUTCD NO.	ITEM 20301-2300 REMOVAL OF SIGN/MARKER (EACH)	ITEM 63308-0000 OBJECT MARKER (EACH)
"MWBR"	103+93	LT	OM-3L	1	1
"MWBR"	103+94	RT	OM-3R	1	1
"MWBR"	114+75	LT	OM-3R	1	1
"MWBR"	114+75	RT	OM-3L	1	1
TOTAL				4	4

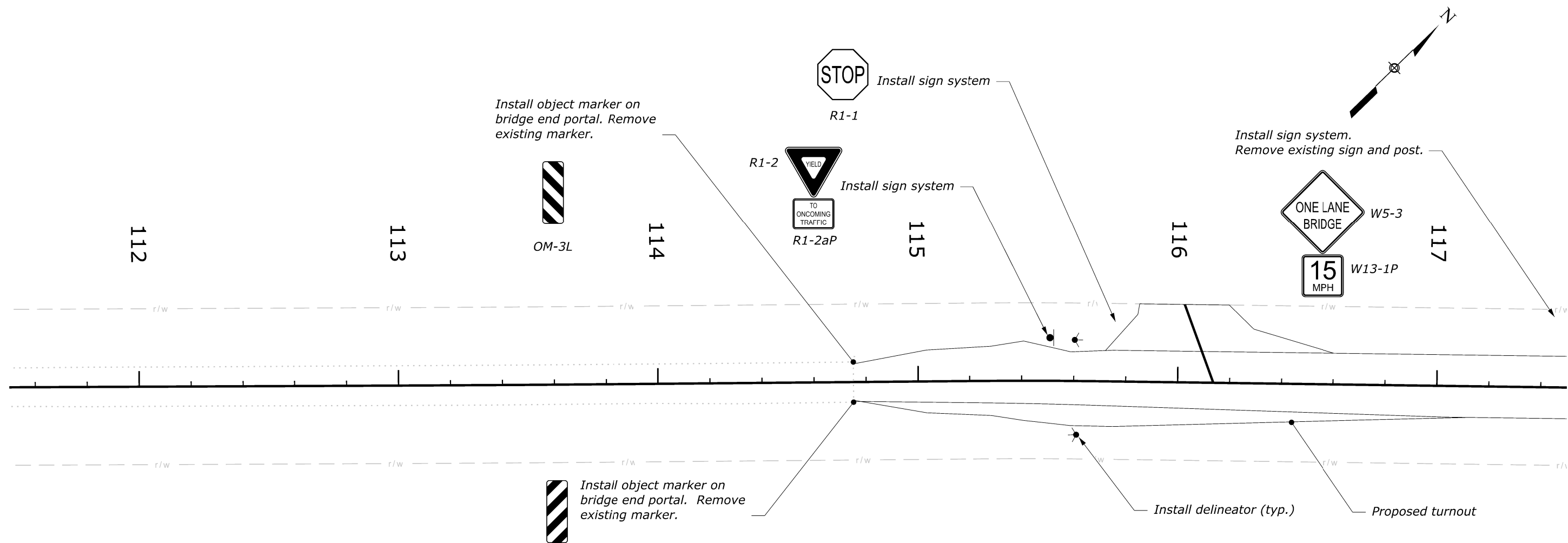
TABULATION OF PERMANENT
TRAFFIC CONTROL QUANTITIES

c:\pw-work\0592538\mt-a-pra\201603_p0.dgn 100+00 TO 105+90 CONTROL PLAN 19 December 2025 4:14 PM

PROJECT	SHEET NUMBER
MT PRAIRIE 2016(3)	K.2

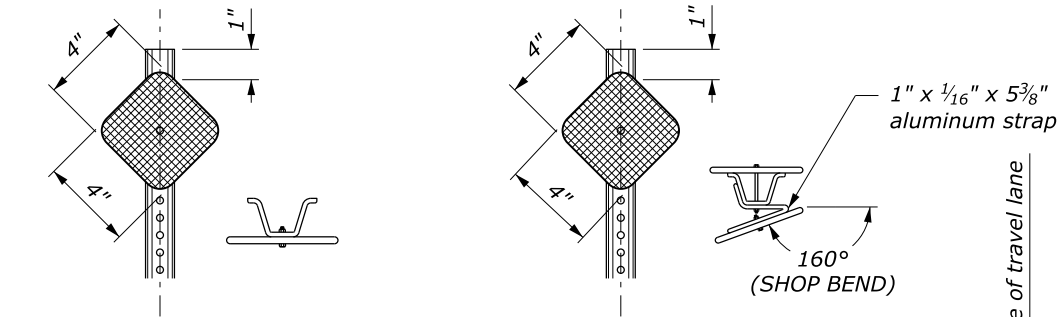


**PERMANENT TRAFFIC
CONTROL PLAN
100+00 TO 105+90**



NOTE:

- When the contract does not include the final surfacing, allow for the thickness of the final pavement structure when establishing the elevation of the traffic delineators.
- Place delineators at a constant clearance distance from the edge of pavement except where guardrail or other obstructions interfere. Align delineators with the inside edge of obstruction. Install delineators located behind beam guardrail so that the delineator post is adjacent to the trailing edge of the nearest guardrail post. (See typical installation with beam type guardrail).
- When a delineator falls within a cross road or approach, the delineator may be moved in either direction a distance not to exceed one quarter of the normal spacing. Eliminate the post if this allowance is exceeded.
- Mount delineators on metal posts with 3/16" cadmium plated bolt(s). Drill or punch a minimum of twelve 3/8" diameter holes on 1-inch centers from the top of the post. 3/8" square holes may be used with large-headed bolt or an appropriate washer. Jam threads after tightening the nut to prevent removal.
- All delineator reflectors have 3/4" corner radii.
- Manufacture posts from flanged U-channel sections of steel meeting the requirements of ASTM A 36 and weighing not less than 1.25 pounds per foot or aluminum meeting the requirements of ASTM B 221, Alloy 6061-T6, with a minimum thickness of 0.125 inches. After fabrication galvanize steel posts in accordance with ASTM A 123.
- When a route has a current ADT of 900 or greater, continuously delineate the roadway along the shoulder by means of post mounted reflectors. Spacing on tangent sections is 400 feet.

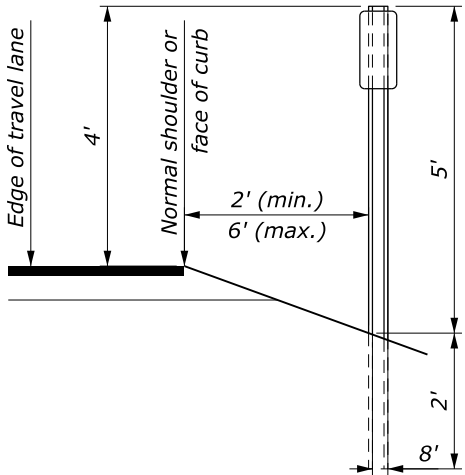


DESIGN A
(WHITE)

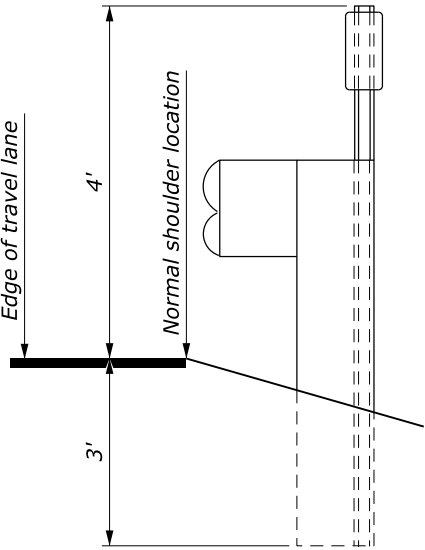
Use for delineation on tangents and on curves with $R > 1500'$.

DESIGN C
(WHITE)

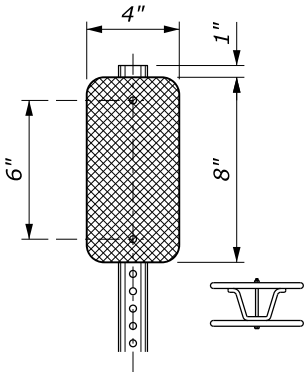
Use on curves with $R \leq 575'$



TYPICAL INSTALLATION

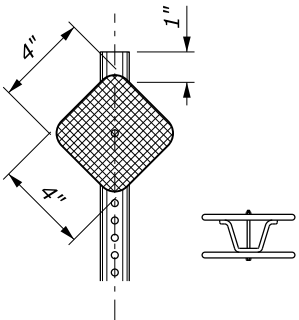


TYPICAL INSTALLATION WITH BEAM TYPE GUARD RAIL



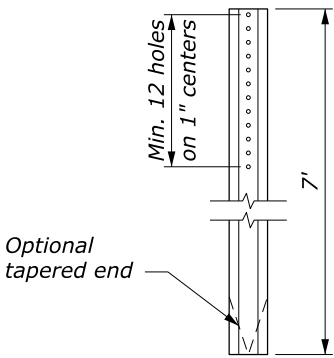
DESIGN D
(YELLOW)

Use at approaches with Stop or Yield signs.



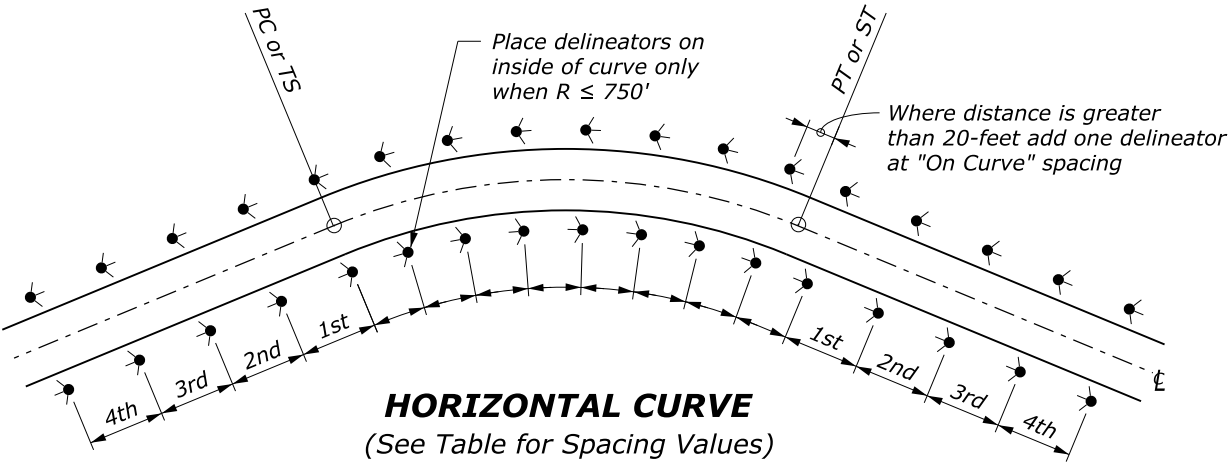
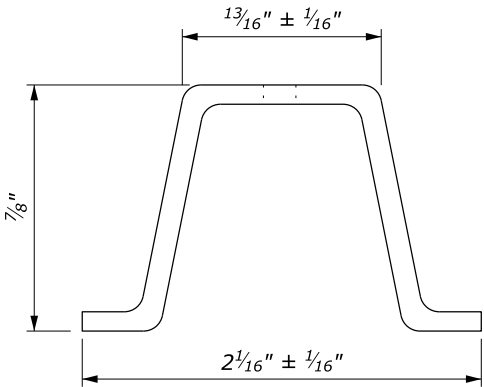
DESIGN F
(WHITE)

Use for curves with $R > 575'$ and $R \leq 1500'$.

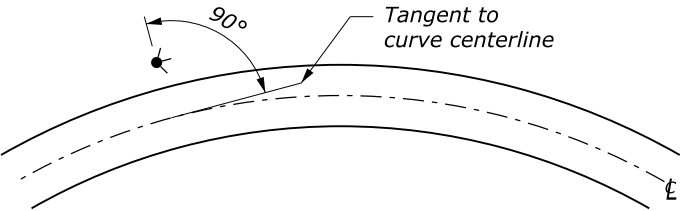


DELINEATOR POST DETAILS

Rigid Steel or Aluminum



HORIZONTAL CURVE
(See Table for Spacing Values)



SIGN INSTALLATION ANGLE

HORIZONTAL CURVES					
RADIUS (ft)	SPACING ON CURVE (ft)	SPACING IN ADVANCE OF & BEYOND CURVE (ft)			
		1st SPACE	2nd SPACE	3rd SPACE	4th SPACE
≥ 5700	300	400	400	400	400
3000 to < 5700	225	400	400	400	400
2000 to < 3000	160	320	400	400	400
1500 to < 2000	130	260	400	400	400
1000 to < 1500	110	220	330	400	400
700 to < 1000	90	185	275	400	400
500 to < 700	75	150	230	300	400
300 to < 500	60	125	185	300	400
< 300	45	90	140	275	400

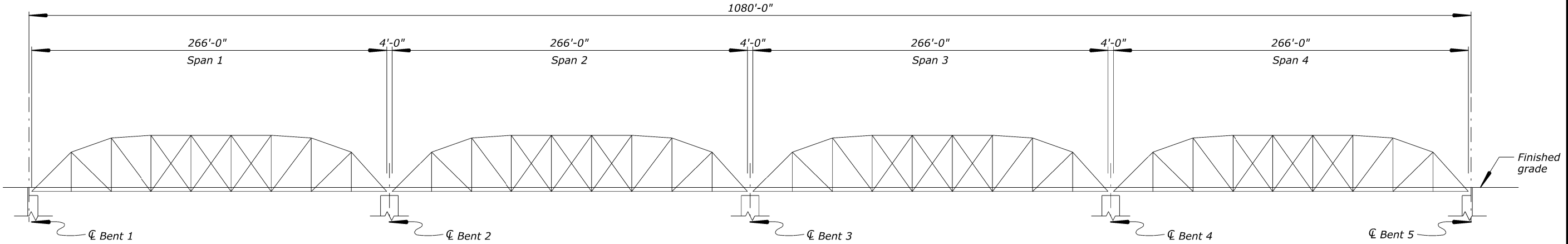
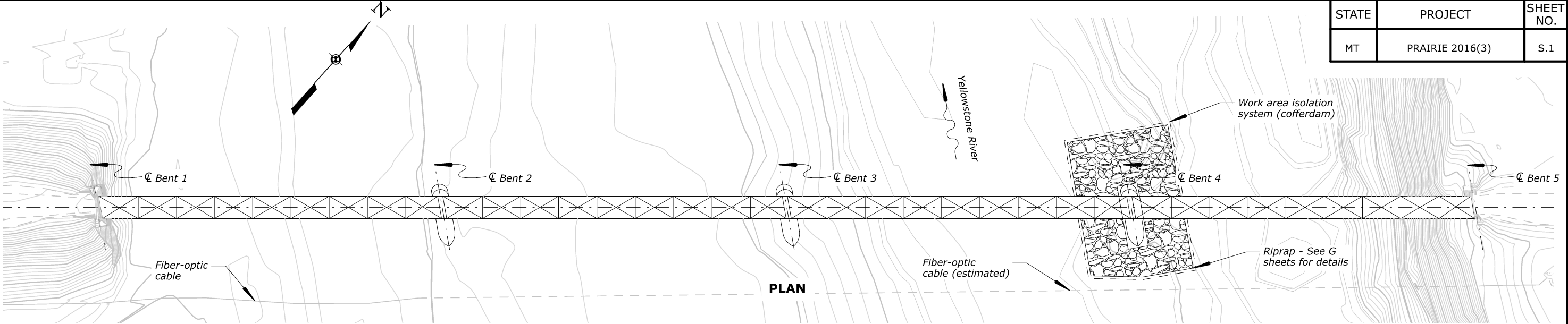
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9 December 2025 4:01 PM

STATE	PROJECT	SHEET NO.
MT	PRAIRIE 2016(3)	S.1



Scope of Work:

1. Install debris and asbestos containment according to Section 203
2. Remove existing bridge rails (W-beam and cable rail)
3. Remove existing deck (steel planks, timber: planks, beams, curb)
4. Clean-up and improve Bents 1 and 5
5. Install proposed steel floorbeam system
6. Install proposed open grid steel deck
7. Install proposed bridge rail
8. Modify Bent 1 and Bent 5 backwalls

PRELIMINARY
NOT FOR CONSTRUCTION

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
WESTERN FEDERAL LANDS HIGHWAY DIVISION

PRAIRIE COUNTY, MT

CALYPSO-MILWAUKEE BRIDGE

PLAN AND ELEVATION

NO.	DATE	BY	REVISIONS	NO.	DATE	BY	REVISIONS	DESIGNED BY	DRAWN BY	CHECKED BY	SCALE	PROJECT TEAM LEADER	BRIDGE DRAWING	DATE	DRAWING NO.
								C. WILLIAMS	B. OLTMANN	O. ELTAYEB	1" = 40'-0"	B. OLTMANN	1 of 13	DECEMBER 2025	RG3317-A

c:\pw-work\00448686\S.1 MT PRAIRIE 2016(3) - Plan and Elev.DGN [Default] 9 December 2025 3:48 PM .DGN ACTUAL FILE: XXXX

GENERAL NOTES:

SPECIFICATIONS:

Design:
AASHTO LRFD Bridge Design Specifications, 9th Edition, 2020.

Construction:
Standard Specifications for Construction of Roads and Bridges on Federal Highway Projects, FP-14.

DESIGN LOADS:

Dead Load:
Concrete: 150 lbs. per cubic foot.
Structural Steel: 490 lbs. per cubic foot.

Live Loads:
HL-93. Maximum Dynamic Load Allowance (Impact), IM=33%, applied to design truck only.

MATERIALS:

Design stresses:

Minor Concrete: $f'c = 4,500$ psi

Reinforcing Steel: $f_y = 60,000$ psi

Cast-in-place Concrete:
Furnish Minor Concrete for bent backwall modifications and fill at metal deck joints.

Reinforcing Steel:
Furnish reinforcing steel conforming to AASHTO M 31, or M 322, Grade 60 deformed. Use minimum concrete cover to the face of any bar of 2 inches, unless noted otherwise on the plans. Minimum splice lengths for all bar sizes are shown on the plans. Provide epoxy coated reinforcing steel, as noted on the bar list.

Structural Steel:
Furnish structural steel, and railing components conforming to ASTM AASHTO M270, Grade 50
Galvanize all structural steel, including floorbeams, open grate metal deck and accessories according to ASTM A123
Furnish bolts, nuts, and threaded rods conforming to ASTM F3125, Grade A325, Type 3, Heavey Hex Style, unless noted.

ESTIMATE OF QUANTITIES				
ITEM NO.	ITEM	QUANTITY	UNIT	NOTES
20302-0150	Removal of bridge rail	2160	LNFT	(1)
20303-0200	Removal of bridge deck	1520	SQYD	(1)
55222-0000	Repair Concrete (Bent Improvement)	ALL	LPSM	(2)
55502-0000	Structural steel, furnished, fabricated, and erected (floorbeams, open grid, misc steel)	382000	LB	(1)
55601-0900	Bridge railing, steel	2170	LNFT	(1)
56202-0000	Temporary Support Structure (work platform)	ALL	LPSM	
56320-0000	Containment system and worker protection	ALL	LPSM	

(1) Contract Quantity
(2) Information Only Quantity:
Minor Concrete = 12 CUYD
Reinforcing Steel = 900 LB

STATE	PROJECT	SHEET NO.
MT	PRAIRIE 2016(3)	S.2

INDEX OF BRIDGE PLAN SHEETS

SHEET NO.	SHEET TITLE
S.1	PLAN AND ELEVATION
S.2	GENERAL NOTES
S.3	TYPICAL SPAN PLAN AND ELEVATION
S.4	TYPICAL SECTIONS
S.5	BENT CLEANUP PLAN
S.6	BENT IMPROVEMENTS
S.7	DECK FRAMING PLAN
S.8	TYPICAL SECTION DETAILS
S.9	DECK PANEL LAYOUT
S.10	DECK PANEL STUD SPACING
S.11	EXPANSTION BOLT SPACING
S.12	RAIL POST DETAILS
S.13	RAIL POST CLEARANCE DETAILS

PRELIMINARY
NOT FOR CONSTRUCTION

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
WESTERN FEDERAL LANDS HIGHWAY DIVISION

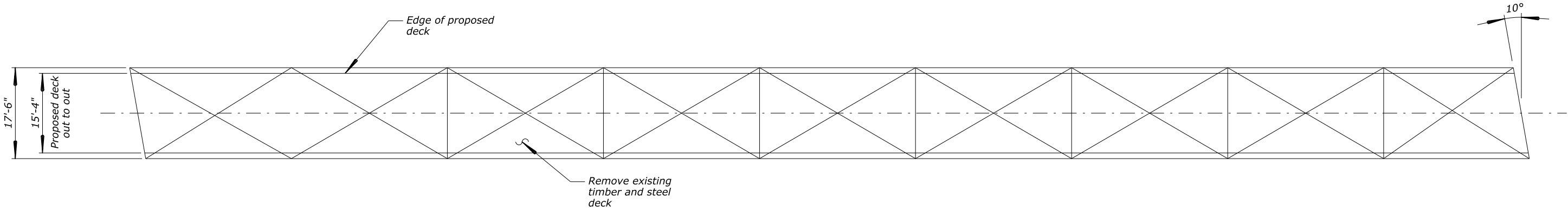
PRAIRIE COUNTY, MT

CALYPSO-MILWAUKEE BRIDGE

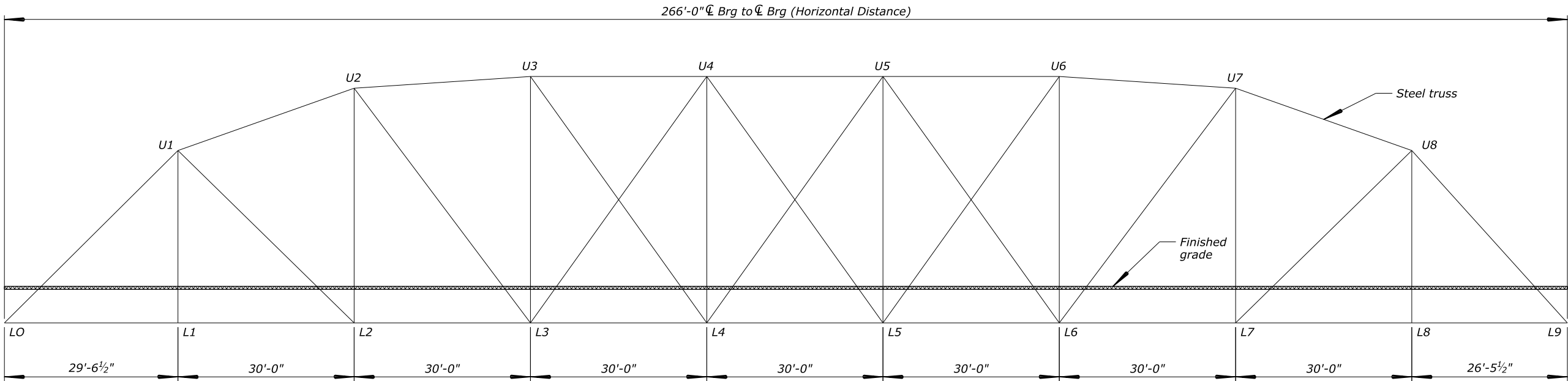
GENERAL NOTES

NO.	DATE	BY	REVISIONS	NO.	DATE	BY	REVISIONS	DESIGNED BY	DRAWN BY	CHECKED BY	SCALE	PROJECT TEAM LEADER	BRIDGE DRAWING	DATE	DRAWING NO.
								C. WILLIAMS	C. WILLIAMS	O. ELTAYEB	NO SCALE	B. OTLMANN	2 of 13	DECEMBER 2025	RG3317-B

STATE	PROJECT	SHEET NO.
MT	PRAIRIE 2016(3)	S.3



PLAN



ELEVATION

PRELIMINARY
NOT FOR CONSTRUCTION

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
WESTERN FEDERAL LANDS HIGHWAY DIVISION

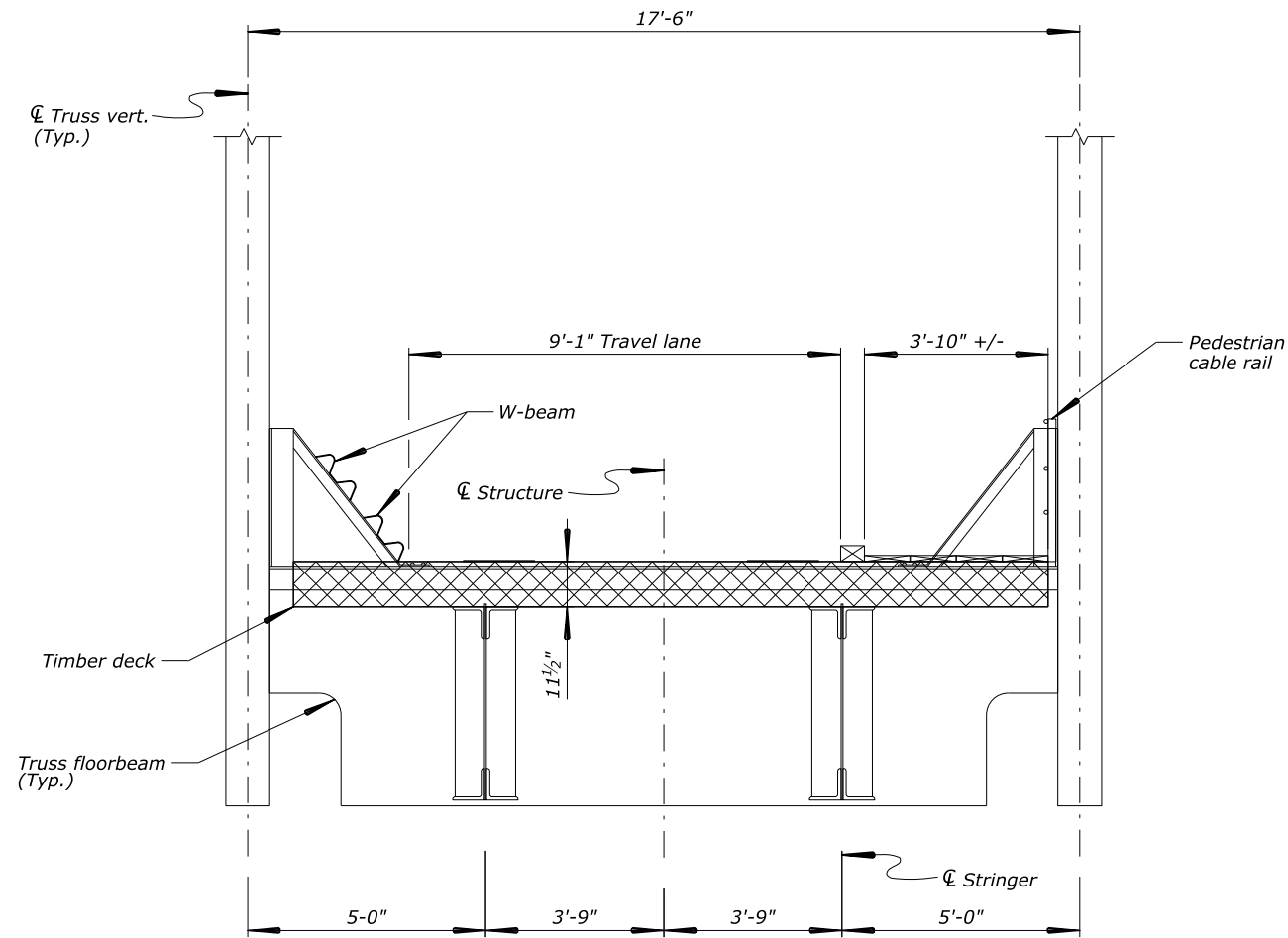
PRAIRIE COUNTY, MT

CALYPSO-MILWAUKEE BRIDGE

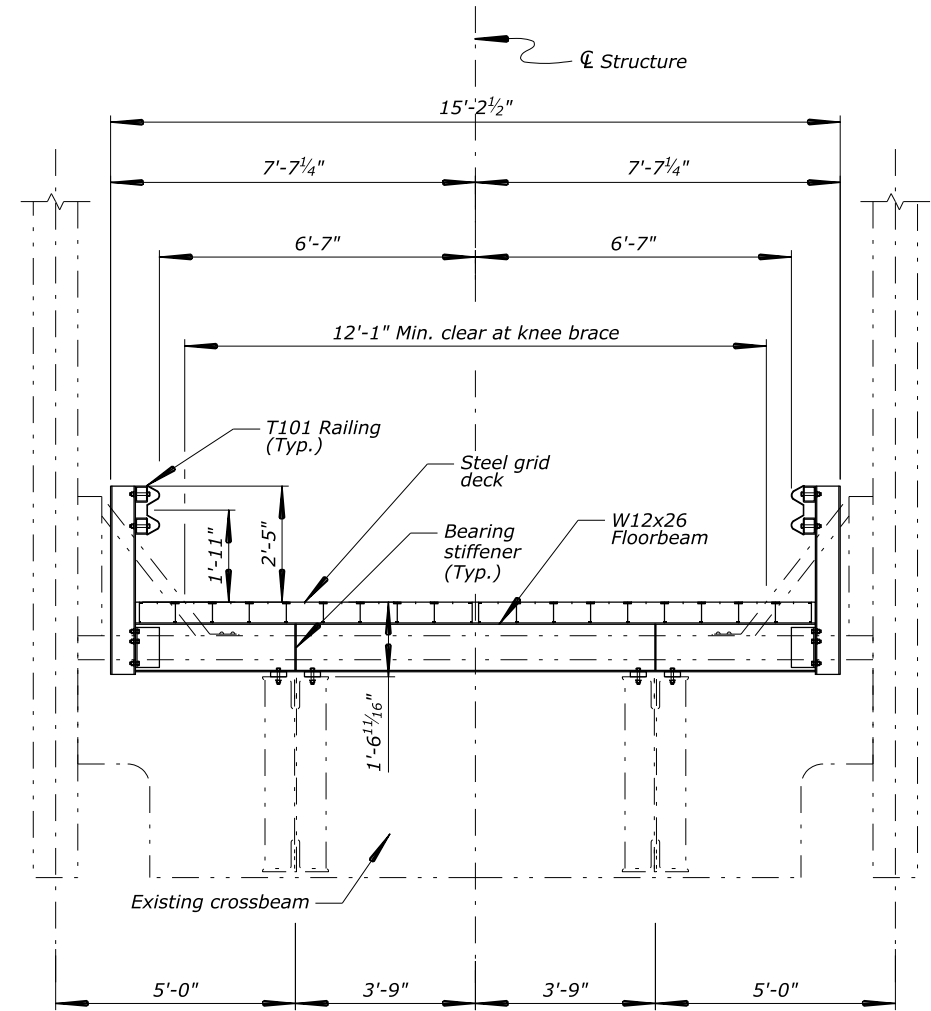
SINGLE SPAN
PLAN AND ELEVATION

NO.	DATE	BY	REVISIONS	NO.	DATE	BY	REVISIONS	DESIGNED BY	DRAWN BY	CHECKED BY	SCALE	PROJECT TEAM LEADER	BRIDGE DRAWING	DATE	DRAWING NO.
								C. WILLIAMS	B. OLTMANN	O. ELTAYEB	1" = 10'-0"	B. OLTMANN	3 of 13	DECEMBER 2025	RG3317-C

STATE	PROJECT	SHEET NO.
MT	PRAIRIE 2016(3)	S.4



TYPICAL EXISTING SECTION



TYPICAL PROPOSED SECTION

NOTES:

1. Remove existing timber deck, w-beam rails, and pedestrian cable rail

PRELIMINARY
NOT FOR CONSTRUCTION

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
WESTERN FEDERAL LANDS HIGHWAY DIVISION

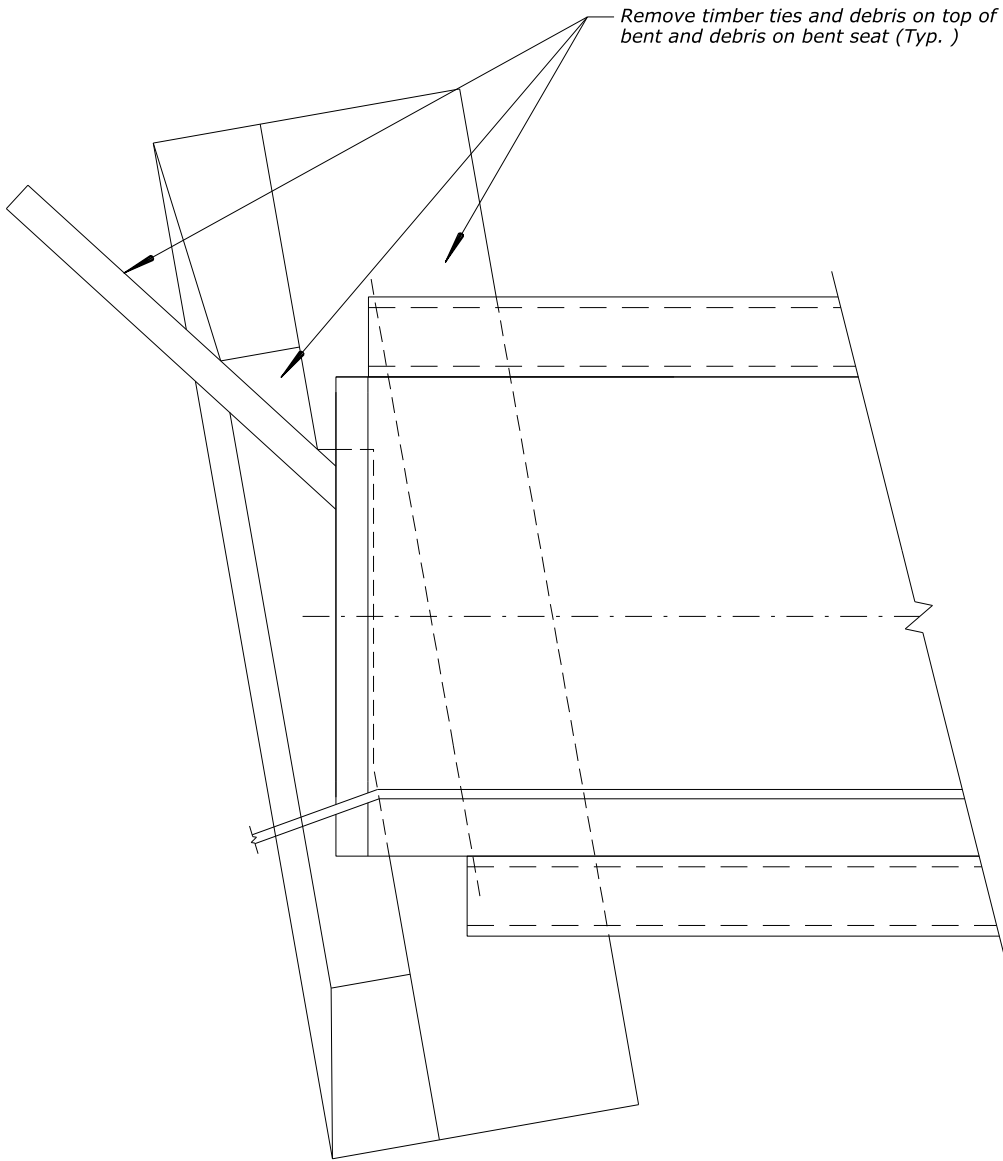
PRAIRIE COUNTY, MT

CALYPSO-MILWAUKEE BRIDGE

TYPICAL SECTION

NO.	DATE	BY	REVISIONS	NO.	DATE	BY	REVISIONS	DESIGNED BY	DRAWN BY	CHECKED BY	SCALE	PROJECT TEAM LEADER	BRIDGE DRAWING	DATE	DRAWING NO.
								C. WILLIAMS	B. Oltmann	O. ELTAYEB	1/2" = 1'-0"	B. OTLMANN	4 of 13	DECEMBER 2025	RG3317-D

STATE	PROJECT	SHEET NO.
MT	PRAIRIE 2016(3)	S.5



BENT CLEANUP PLAN

(Bent 1 shown, bent 5 similar)



TYPICAL BENT

(Bent 1 shown, bent 5 similar)

Note:
Remove all debris and obstructions to provide a clean, debris-free bent seat at Bents 1 & 5.

PRELIMINARY
NOT FOR CONSTRUCTION

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
WESTERN FEDERAL LANDS HIGHWAY DIVISION

PRAIRIE COUNTY, MT

CALYPSO-MILWAUKEE BRIDGE

BENT CLEANUP PLAN

NO.	DATE	BY	REVISIONS	NO.	DATE	BY	REVISIONS	DESIGNED BY	DRAWN BY	CHECKED BY	SCALE	PROJECT TEAM LEADER	BRIDGE DRAWING	DATE	DRAWING NO.
								C. WILLIAMS	C. WILLIAMS	O. ELTAYEB	NO SCALE	B. OTLMANN	5 of 13	DECEMBER 2025	RG3317-E

ACTUAL FILE: XXXX .DGN

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9 December 2025 4:09 PM

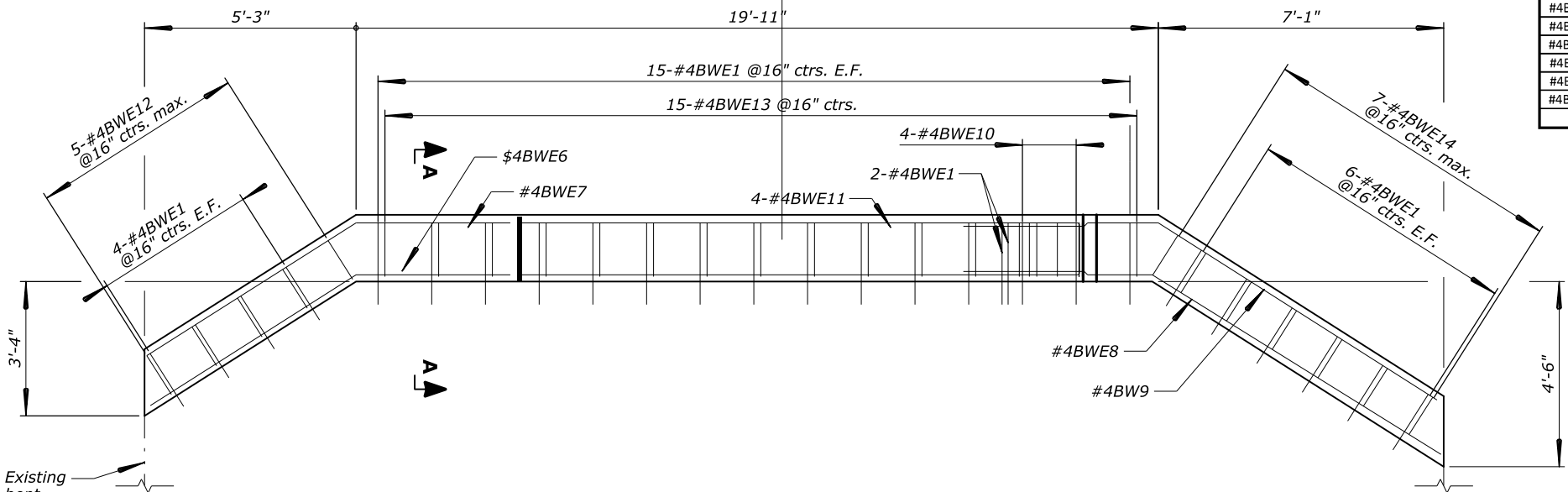
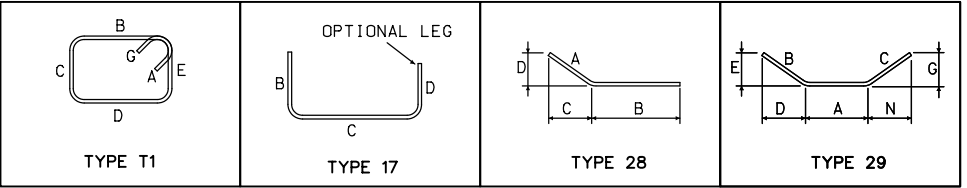
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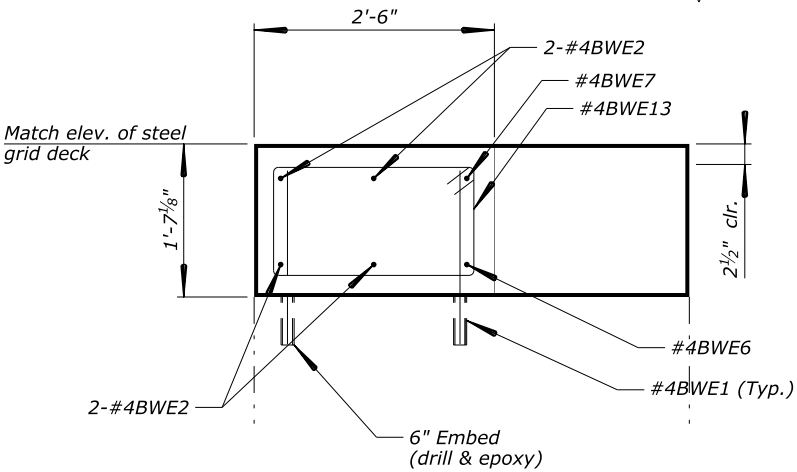
9 December 2025 3:34 PM

STATE	PROJECT	SHEET NO.
MT	PRAIRIE 2016(3)	S.6

REINFORCING STEEL SCHEDULE						DIMENSION TABLE							
BENT 1													
BAR MK	SIZE	TYPE	QTY	LENGTH	WEIGHT	A	B	C	D	E	G	N	
#4BWE1A	4	STR	20	1-7	21.2								
#4BWE1B	4	STR	32	1-10	39.2								
#4BWE2	4	29	2	34-1	45.5	19-11	6-0	8-2	5-1	3-2 1/2"	4-4 1/2"	6-11	
#4BWE3	4	29	2	34-1	45.5	19-9	6-1	8-3	5-2	3-3	4-5	6-11 1/2"	
#4BWE4	4	STR	2	5-8	7.6								
#4BWE5	4	STR	2	7-7	10.1								
#4BWE6	4	28	1	10-0	6.7	6-0	4-0	5-1	3-2 1/2"				
#4BWE7	4	28	1	10-0	6.7	6-1	3-11	5-2	3-3				
#4BWE8	4	28	1	13-0	9.2	8-2	4-10	6-11	4-4 1/2"				
#4BWE9	4	28	1	13-0	8.7	8-3	4-9	6-11 1/2"	4-5				
#4BWE10	4	17	4	8-10	23.6		1-2	3-10	3-10				
#4BWE11	4	17	4	29-1	77.7		1-1	14-0	14-0				
#4BWE12A	4	T1	1	8-8	5.8	4 1/2"	3-0	1-1 1/2"	3-0	1-1 1/2"	4 1/2"		
#4BWE12B	4	T1	1	8-2	5.5	4 1/2"	2-9	1-1 1/2"	2-9	1-1 1/2"	4 1/2"		
#4BWE12C	4	T1	1	7-10	5.2	4 1/2"	2-7	1-1 1/2"	2-7	1-1 1/2"	4 1/2"		
#4BWE12D	4	T1	1	7-4	4.9	4 1/2"	2-4	1-1 1/2"	2-4	1-1 1/2"	4 1/2"		
#4BWE12E	4	T1	1	7-0	4.7	4 1/2"	2-2	1-1 1/2"	2-2	1-1 1/2"	4 1/2"		
#4BWE13A	4	T1	6	7-5	29.7	4 1/2"	2-2	1-2	2-2	1-2	4 1/2"		
#4BWE13B	4	T1	1	6-9	4.5	4 1/2"	1-10	1-2	1-10	1-2	4 1/2"		
#4BWE13C	4	T1	1	7-3	4.8	4 1/2"	2-1	1-2	2-1	1-2	4 1/2"		
#4BWE13D	4	T1	1	7-7	5.1	4 1/2"	2-3	1-2	2-3	1-2	4 1/2"		
#4BWE13E	4	T1	1	8-1	5.4	4 1/2"	2-6	1-2	2-6	1-2	4 1/2"		
#4BWE13F	4	T1	1	8-7	5.7	4 1/2"	2-9	1-2	2-9	1-2	4 1/2"		
#4BWE13G	4	T1	1	9-1	6.1	4 1/2"	3-0	1-2	3-0	1-2	4 1/2"		
#4BWE13H	4	T1	1	9-7	6.4	4 1/2"	3-3	1-2	3-3	1-2	4 1/2"		
#4BWE13I	4	T1	1	10-1	6.7	4 1/2"	3-6	1-2	3-6	1-2	4 1/2"		
#4BWE13J	4	T1	1	10-5	7.0	4 1/2"	3-8	1-2	3-8	1-2	4 1/2"		
#4BWE14A	4	T1	1	7-0	4.7	4 1/2"	2-2	1-1 1/2"	2-2	1-1 1/2"	4 1/2"		
#4BWE14B	4	T1	1	7-2	4.8	4 1/2"	2-3	1-1 1/2"	2-3	1-1 1/2"	4 1/2"		
#4BWE14C	4	T1	1	7-4	4.9	4 1/2"	2-4	1-1 1/2"	2-4	1-1 1/2"	4 1/2"		
#4BWE14D	4	T1	1	7-6	5.0	4 1/2"	2-5	1-1 1/2"	2-5	1-1 1/2"	4 1/2"		
#4BWE14E	4	T1	1	7-10	5.2	4 1/2"	2-7	1-1 1/2"	2-7	1-1 1/2"	4 1/2"		
#4BWE14F	4	T1	1	8-0	5.3	4 1/2"	2-8	1-1 1/2"	2-8	1-1 1/2"	4 1/2"		
#4BWE14G	4	T1	1	8-4	5.6	4 1/2"	2-10	1-1 1/2"	2-10	1-1 1/2"	4 1/2"		
INFORMATION ONLY SUBTOTAL (EPOXY)					450.0								



BENT 1 ELEVATION



SECTION A

NOTE:

- Bent 1 shown, similar improvement plan for Bent 5.
- The suffix E denotes epoxy coated reinforcement

KEY:

E.F. = Each Face
Top = Top
Bott. = Bottom
BWE = Backwall Epoxy Bar

PRELIMINARY
NOT FOR CONSTRUCTION

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
WESTERN FEDERAL LANDS HIGHWAY DIVISION

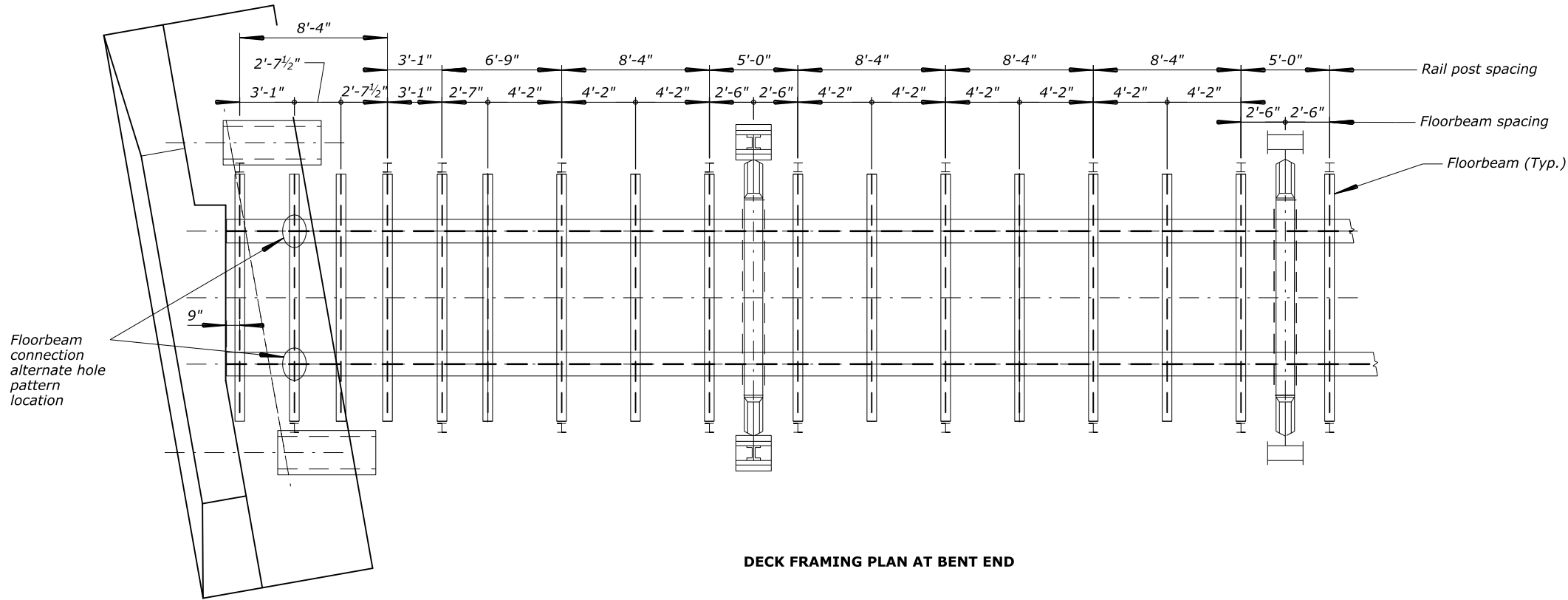
PRAIRIE COUNTY, MT

CALYPSO-MILWAUKEE BRIDGE

BENT IMPROVEMENTS

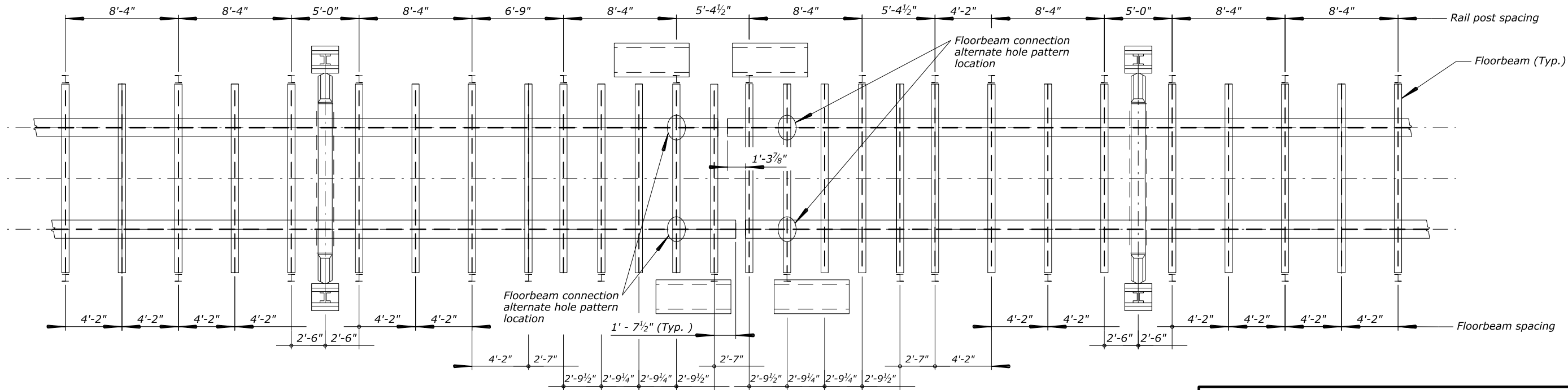
NO.	DATE	BY	REVISIONS	NO.	DATE	BY	REVISIONS	DESIGNED BY	DRAWN BY	CHECKED BY	SCALE	PROJECT TEAM LEADER	BRIDGE DRAWING	DATE	DRAWING NO.
								C. WILLIAMS	C. WILLIAMS	O. ELTAYEB	1/2" = 1'-0"	B. OTLMANN	6 of 13	DECEMBER 2025	RG3317-F

STATE	PROJECT	SHEET NO.
MT	PRAIRIE 2016(3)	S.7



FLOORBEAM ESTIMATED WEIGHTS - FOR INFORMATION ONLY		
No. of Floorbeam	Weight Each (LB)	Total Floorbeam Weight (LB)
124	369	45728

DECK FRAMING PLAN AT BENT END



DECK FRAMING PLAN AT PIER

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WESTERN FEDERAL LANDS HIGHWAY DIVISION

PRAIRIE COUNTY, MT

CALYPSO-MILWAUKEE BRIDGE

DECK FRAMING LAYOUT

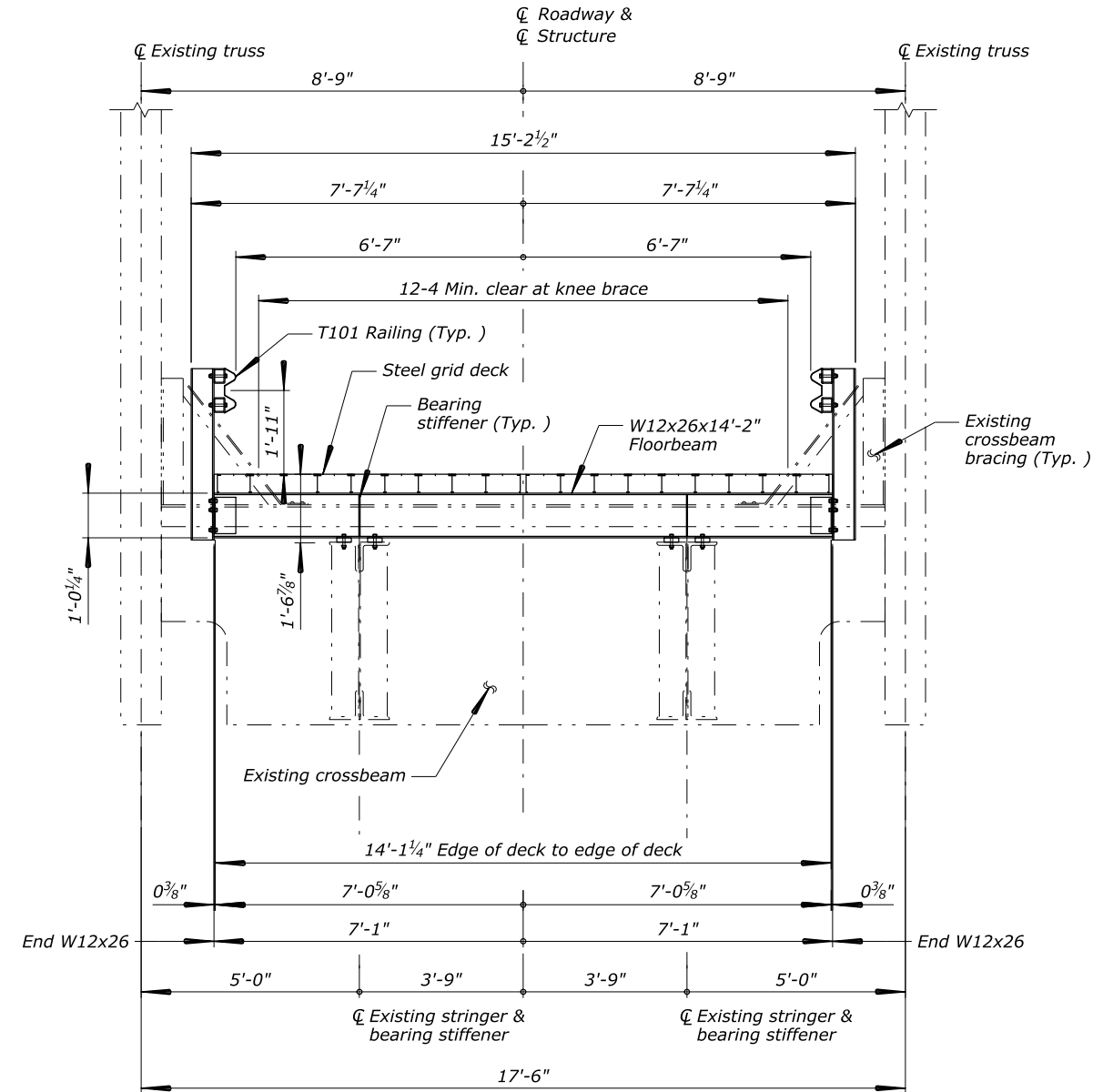
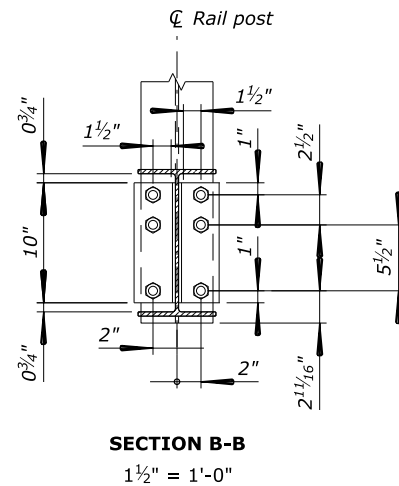
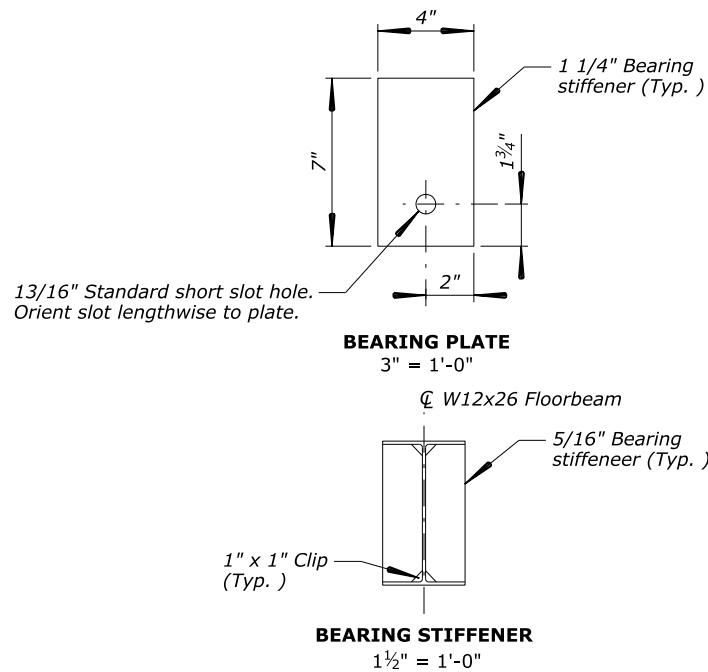
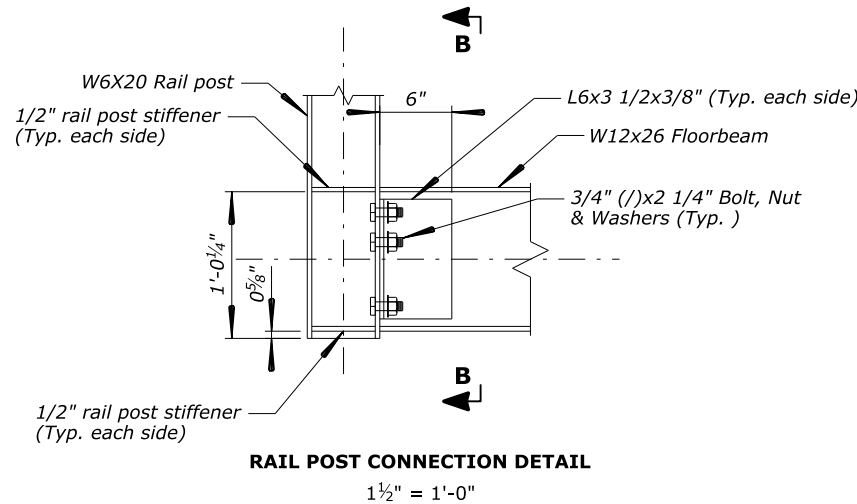
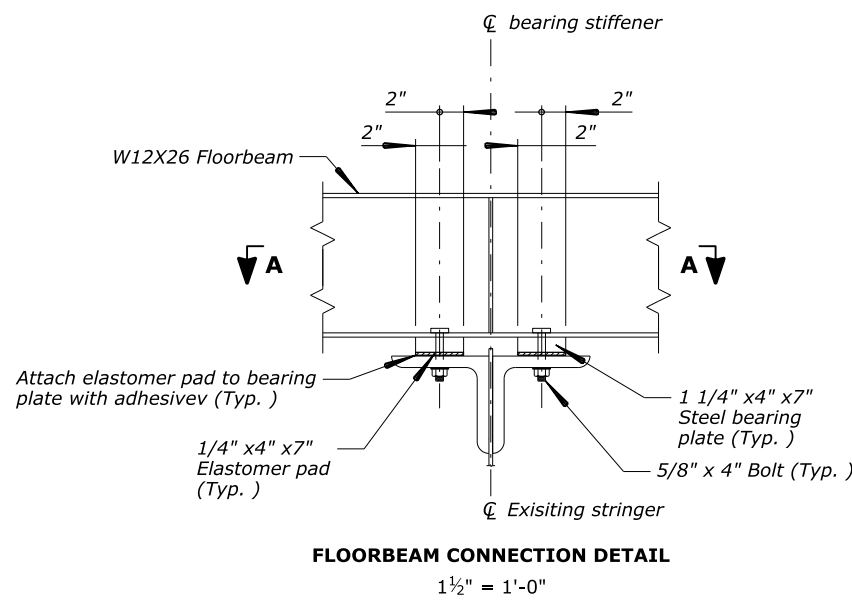
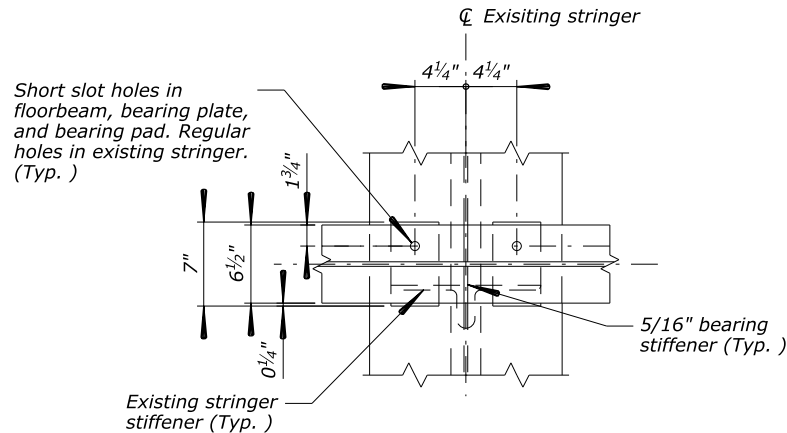
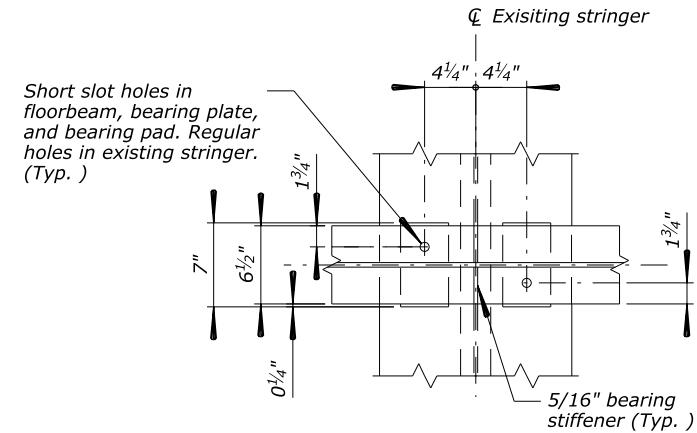
PRELIMINARY
NOT FOR CONSTRUCTION

NO.	DATE	BY	REVISIONS	NO.	DATE	BY	REVISIONS	DESIGNED BY	DRAWN BY	CHECKED BY	SCALE	PROJECT TEAM LEADER	BRIDGE DRAWING	DATE	DRAWING NO.
								C. WILLIAMS	C. WILLIAMS	O. ELTAYEB	1/4" = 1'-0"	B. OTLMANN	7 of 13	DECEMBER 2025	RG3317-G

ACTUAL FILE: XXXX .DGN

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9 December 2025 3:49 PM



PRELIMINARY
NOT FOR CONSTRUCTION

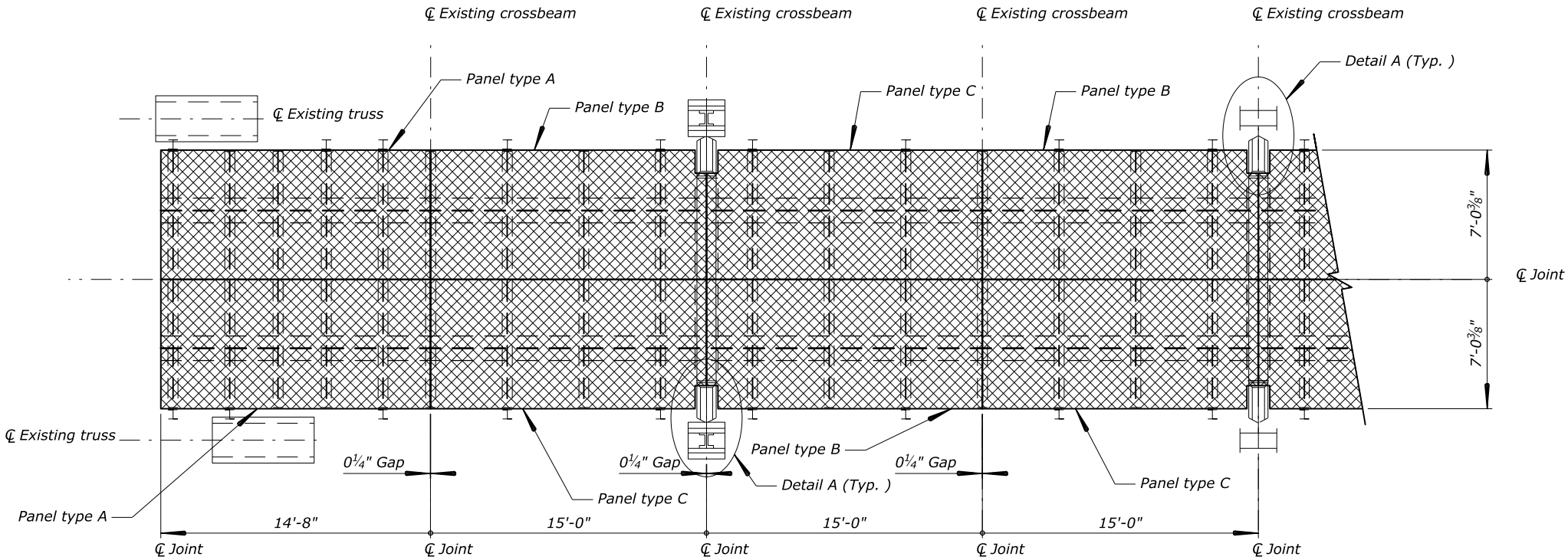
U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
WESTERN FEDERAL LANDS HIGHWAY DIVISION
PRAIRIE COUNTY, MT
CALYPSO-MILWAUKEE BRIDGE
TYPICAL SECTION DETAILS

NO.	DATE	BY	REVISIONS	NO.	DATE	BY	REVISIONS	DESIGNED BY	DRAWN BY	CHECKED BY	SCALE	PROJECT TEAM LEADER	BRIDGE DRAWING	DATE	DRAWING NO.
								C. WILLIAMS	C. WILLIAMS	O. ELTAYEB	AS SHOWN	B. OTLMANN	8 of 13	DECEMBER 2025	RG3317-H

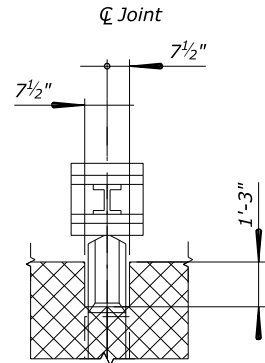
ACTUAL FILE: XXXX .DGN

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SUGGESTED DECK PANEL LAYOUT AT BENTS
1/2" = 1'-0"

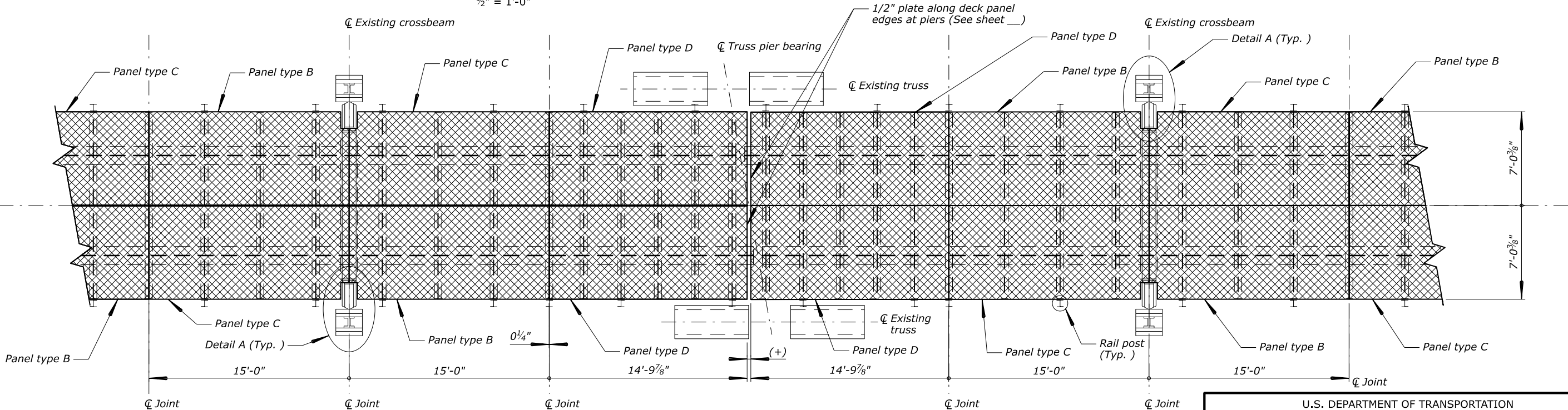


DETAIL A
3/8" = 1'-0"

STEEL GRID DECK ESTIMATED WEIGHTS - FOR INFORMATION ONLY

Panel Type	Panel Area (SQFT)	Panel Weight (LB)	Number of Panels	Total Panel Area (SQFT)	Total Panel Weight (LB)
A	103	2063	4	413	8250
B	105	2094	64	6700	134000
C	105	2094	64	6700	134000
D	104	2084	12	1251	25014
Total	N/A	N/A	144	15063	301264

NOTE: Estimated steel grid deck weight is based on 20 pounds per square foot of deck area. The weight is an estimate only



SUGGESTED DECK PANELS AT PIERS
1/2" = 1'-0"

PRELIMINARY
NOT FOR CONSTRUCTION

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
WESTERN FEDERAL LANDS HIGHWAY DIVISION

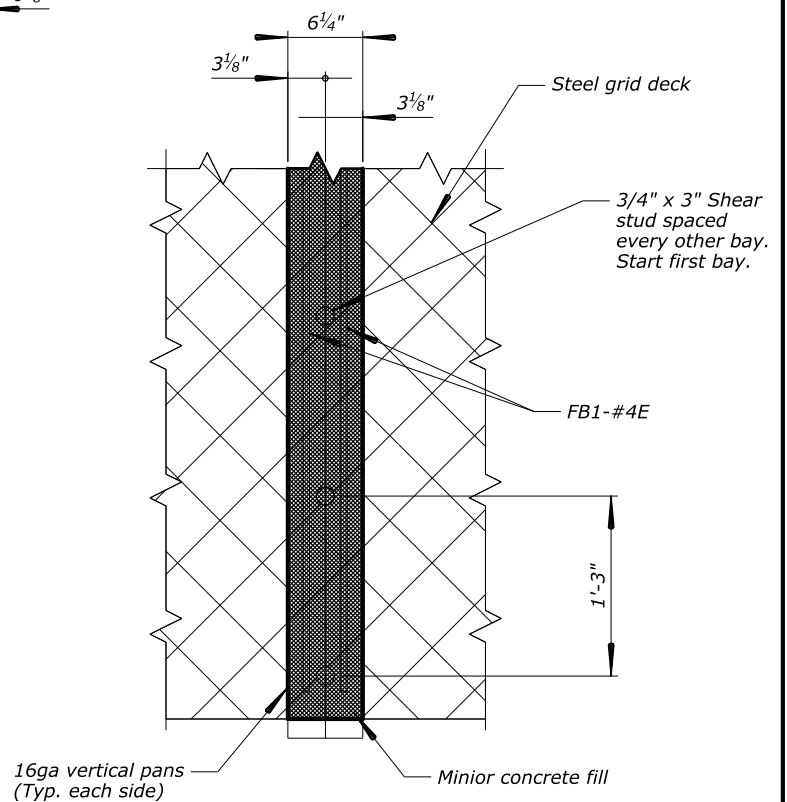
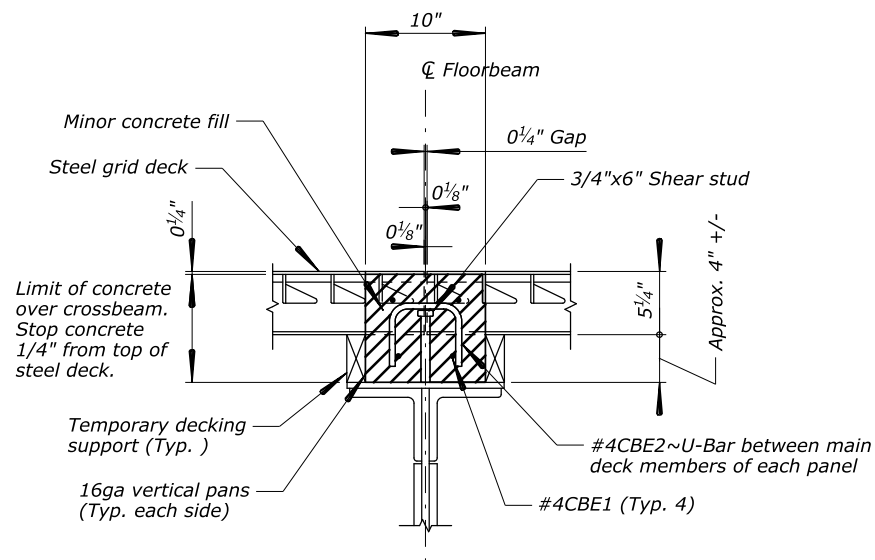
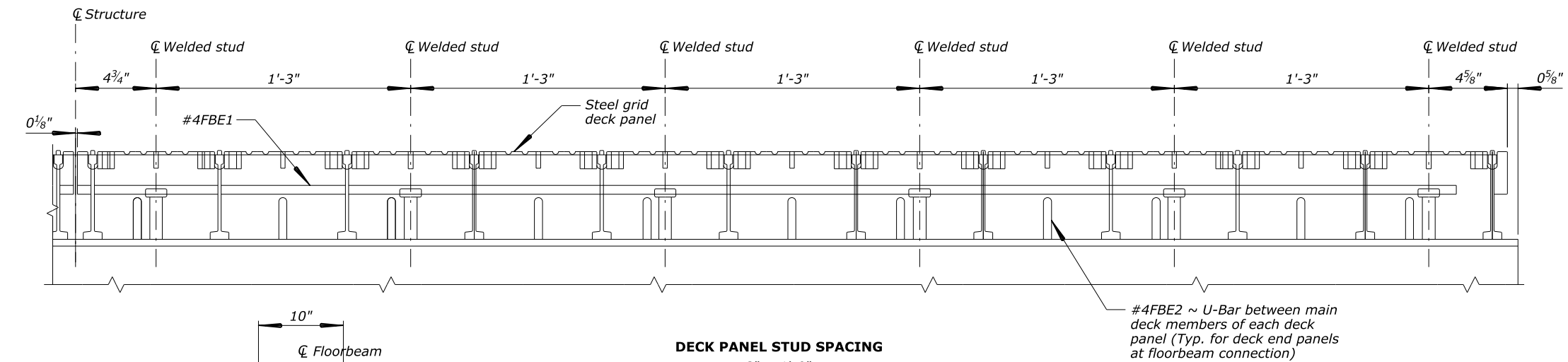
PRAIRIE COUNTY, MT

CALYPSO-MILWAUKEE BRIDGE

DECK PANEL LAYOUT

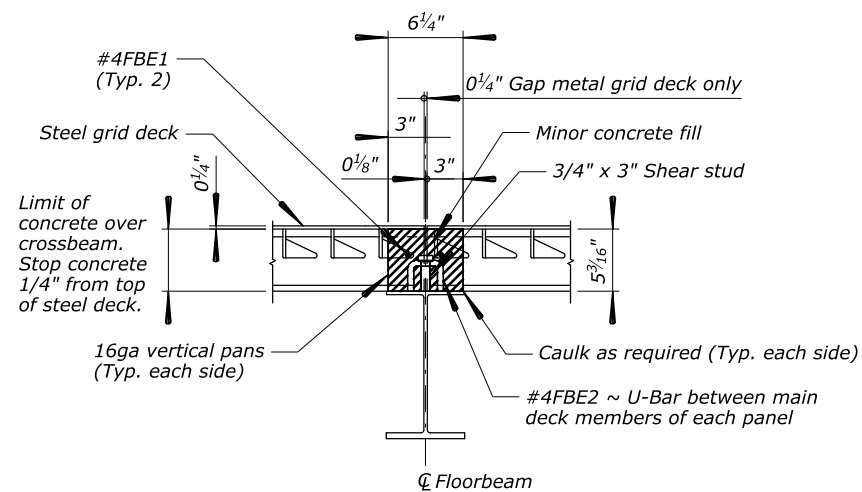
NO.	DATE	BY	REVISIONS	NO.	DATE	BY	REVISIONS	DESIGNED BY	DRAWN BY	CHECKED BY	SCALE	PROJECT TEAM LEADER	BRIDGE DRAWING	DATE	DRAWING NO.
								C. WILLIAMS	C. WILLIAMS	O. ELTAYEB	AS SHOWN	B. OTLMANN	9 of 13	DECEMBER 2025	RG3317-I

STATE	PROJECT	SHEET NO.
MT	PRAIRIE 2016(3)	S.10

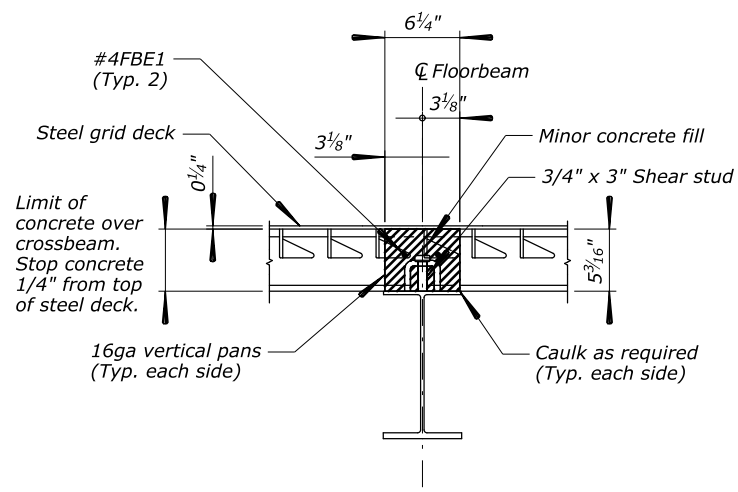


PLAN AT FLOORBEAM CONNECETION
1½" = 1'-0"

DECK PANEL AT EXISTING CROSSBEAM CONNECTION
 $1\frac{1}{2}" = 1'-0"$



DECK END PANELS AT FLOORBEAM CONNECTION
 $1\frac{1}{2}" = 1'-0"$



DECK PANEL AT FLOORBEAM CONNECTION
 $1\frac{1}{2}" = 1'-0"$

REINFORCING STEEL BAR MARK	
BAR MARK	LOCATION
CB	Crossbeam
FB	Floorbeam

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
WESTERN FEDERAL LANDS HIGHWAY DIVISION

PRAIRIE COUNTY, MT

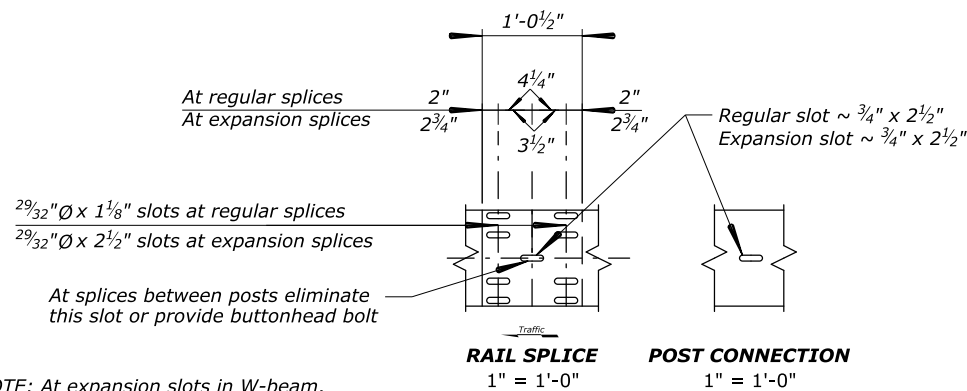
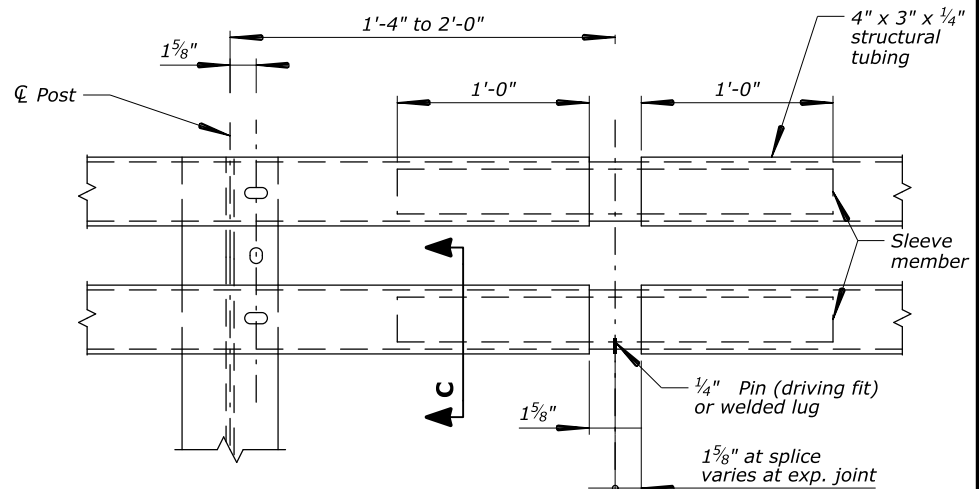
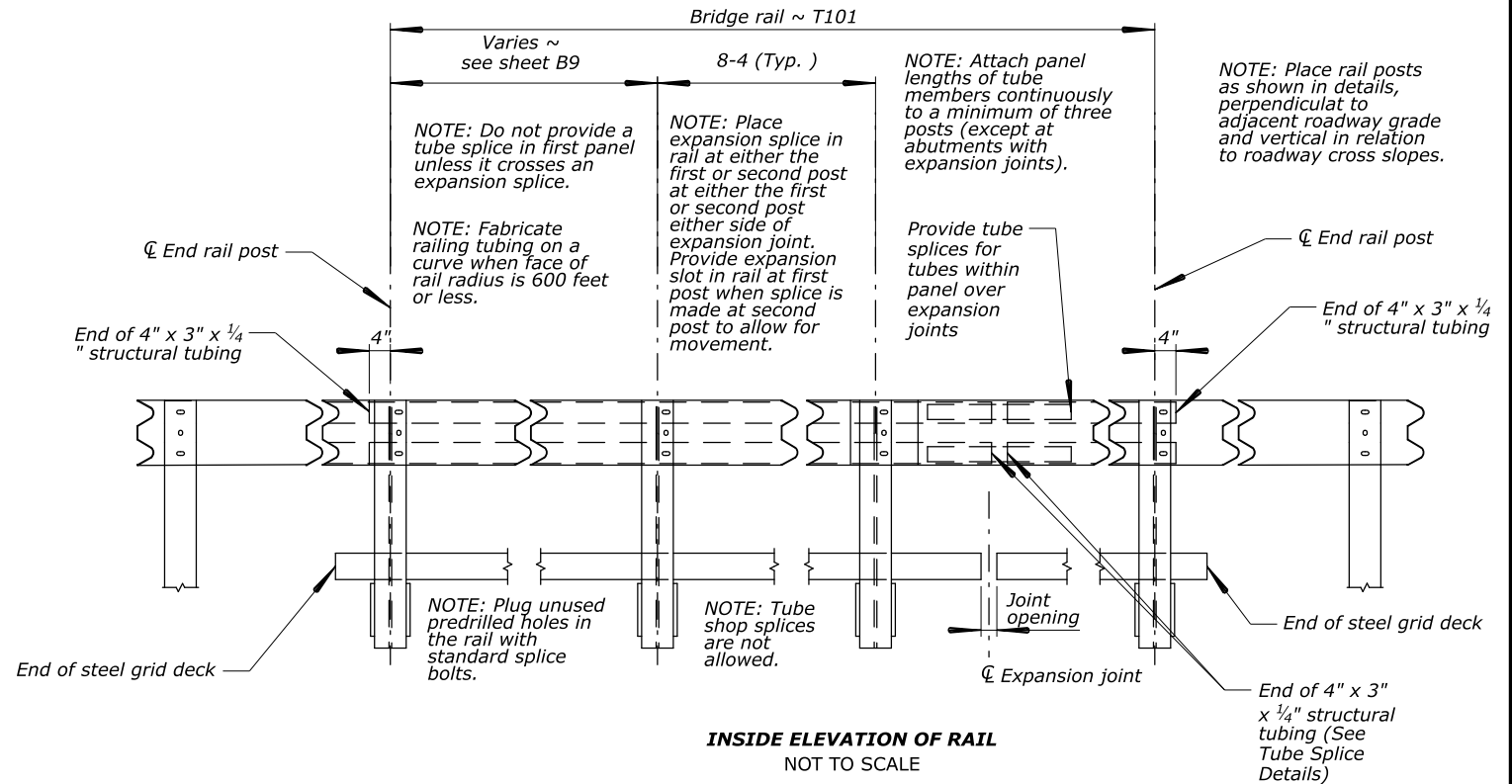
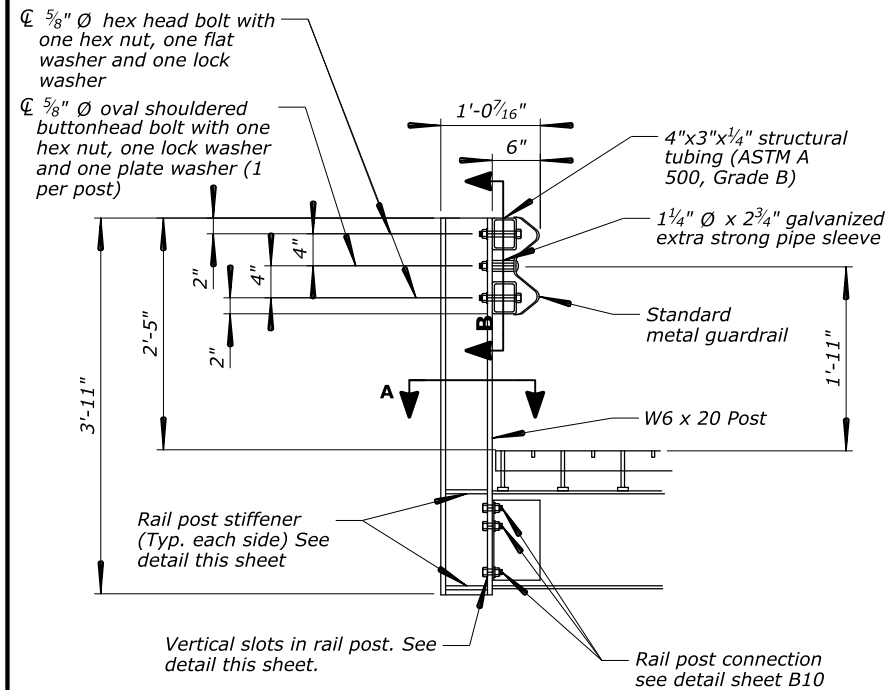
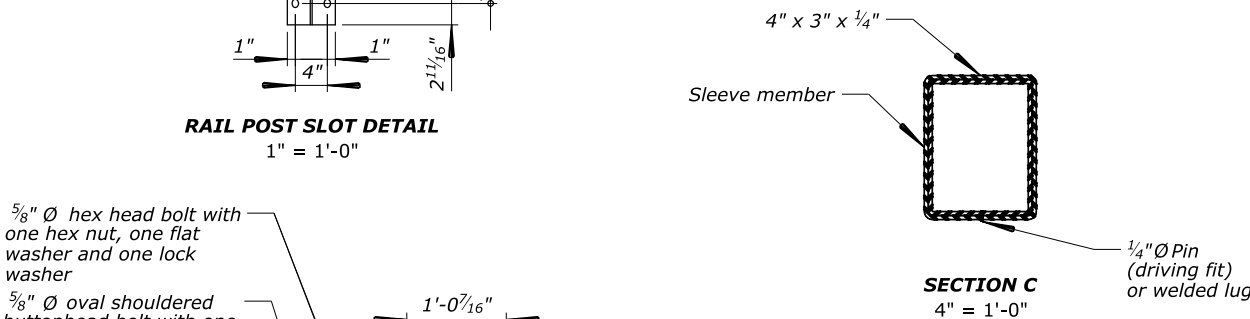
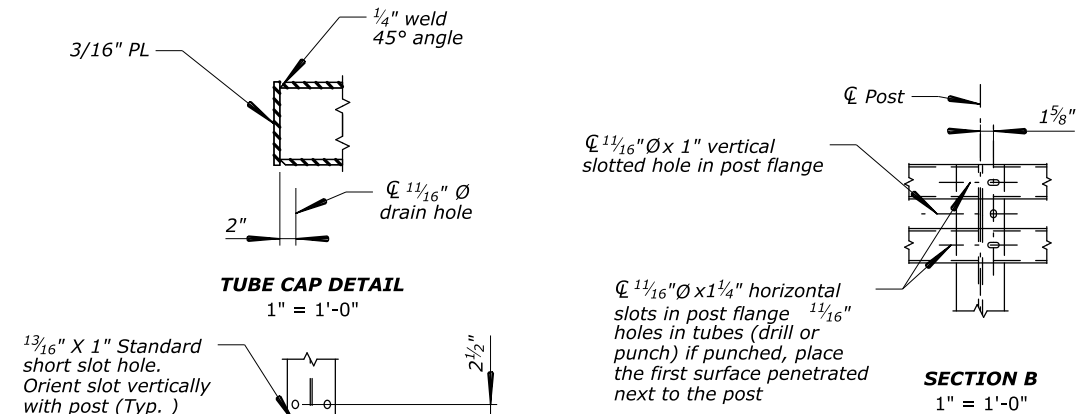
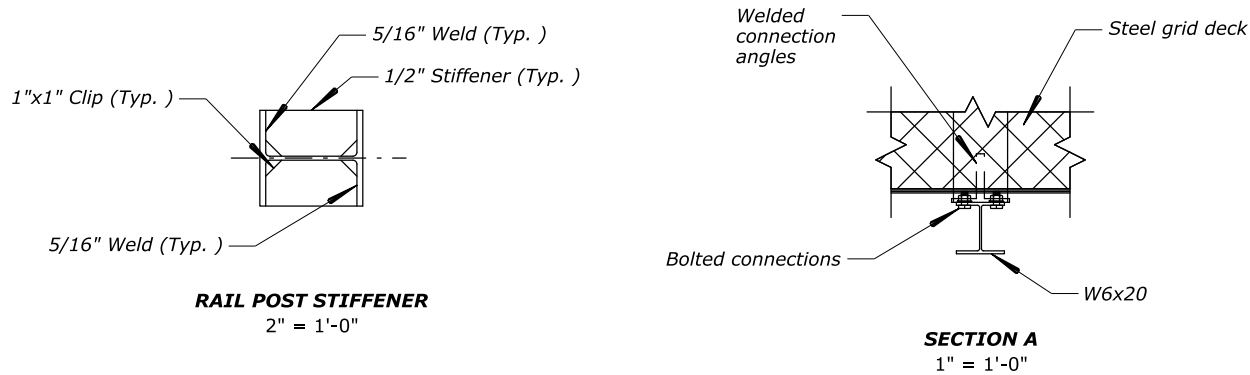
CALYPSO-MILWAUKEE BRIDGE

DECK PANEL STUD SPACING

PRELIMINARY
NOT FOR CONSTRUCTION

NO.	DATE	BY	REVISIONS	NO.	DATE	BY	REVISIONS	DESIGNED BY	DRAWN BY	CHECKED BY	SCALE	PROJECT TEAM LEADER	BRIDGE DRAWING	DATE	DRAWING NO.
								C. WILLIAMS	C. WILLIAMS	O. ELTAYEB	AS SHOWN	B. OTLMANN	10 of 13	DECEMBER 2025	RG3317-J

STATE	PROJECT	SHEET NO.
MT	PRAIRIE 2016(3)	S.12



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FEDERAL HIGHWAY ADMINISTRATION
WESTERN FEDERAL LANDS HIGHWAY DIVISION

PRAIRIE COUNTY, MT

CALYPSO-MILWAUKEE BRIDGE

RAIL POST DETAILS

ACTUAL FILE: XXXX .DGN

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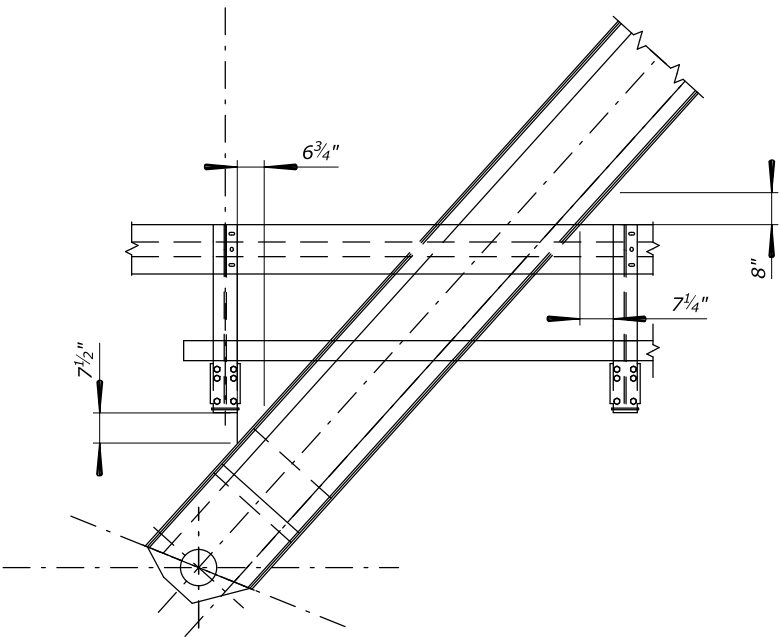
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								C. WILLIAMS	C. WILLIAMS	O. ELTAYEB	AS SHOWN	B. OTLMANN	12 of 13	DECEMBER 2025	RG3317-L

STATE	PROJECT	SHEET NO.
MT	PRAIRIE 2016(3)	S.13

ACTUAL FILE: XXXX .DGN

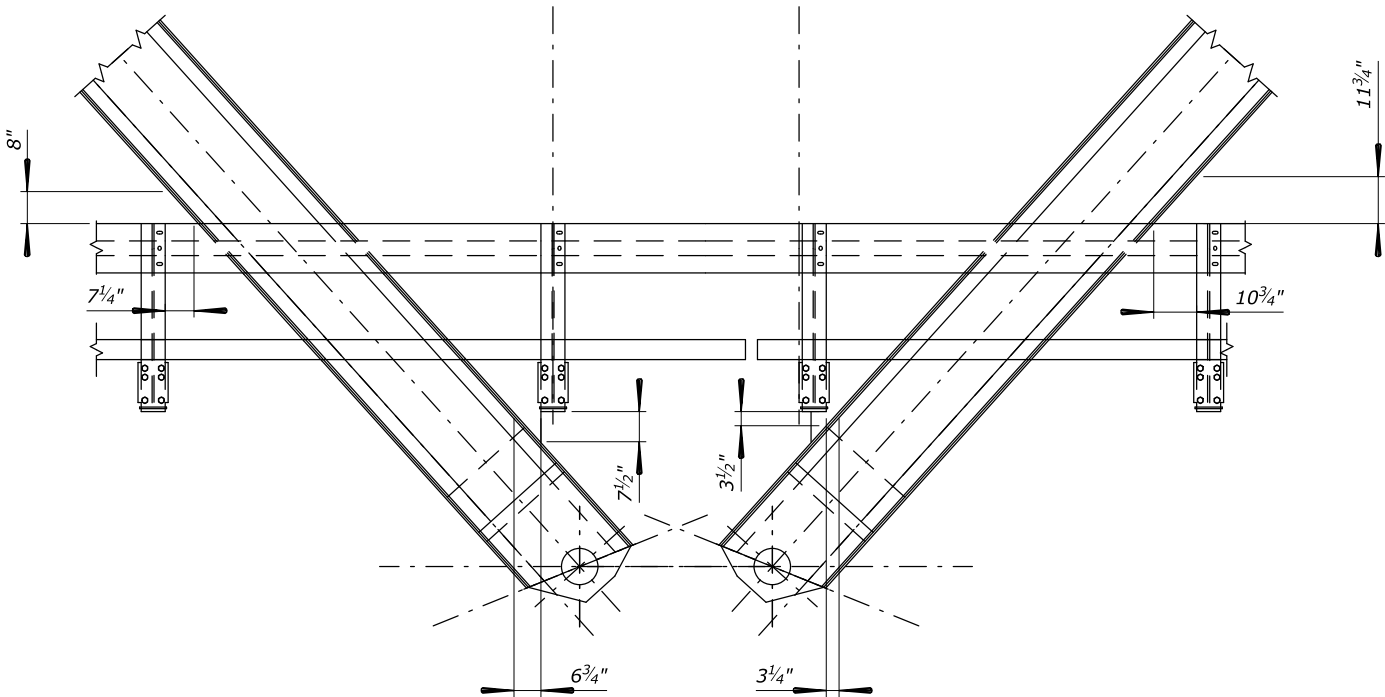
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VIEW A - FOR INFORMATION ONLY

NOTE: This view detail is for information only. This detail shows the intended placement of the T101 rail posts and associated floorbeams relative to the truss diagonal members at the bents.



VIEW B - FOR INFORMATION ONLY

NOTE: This view detail is for information only. This detail shows the intended placement of the T101 rail posts and associated floorbeams relative to the truss diagonal members at the piers.

PRELIMINARY
NOT FOR CONSTRUCTION

U.S. DEPARTMENT OF TRANSPORTATION
FEDERAL HIGHWAY ADMINISTRATION
WESTERN FEDERAL LANDS HIGHWAY DIVISION

PRAIRIE COUNTY, MT

CALYPSO-MILWAUKEE BRIDGE

RAIL POST CLEARANCE DETAILS

NO.	DATE	BY	REVISIONS	NO.	DATE	BY	REVISIONS	DESIGNED BY	DRAWN BY	CHECKED BY	SCALE	PROJECT TEAM LEADER	BRIDGE DRAWING	DATE	DRAWING NO.
								C. WILLIAMS	C. WILLIAMS	O. ELTAYEB	1/2" = 1'-0"	B. OTLMANN	13 of 13	DECEMBER 2025	RG3317-M